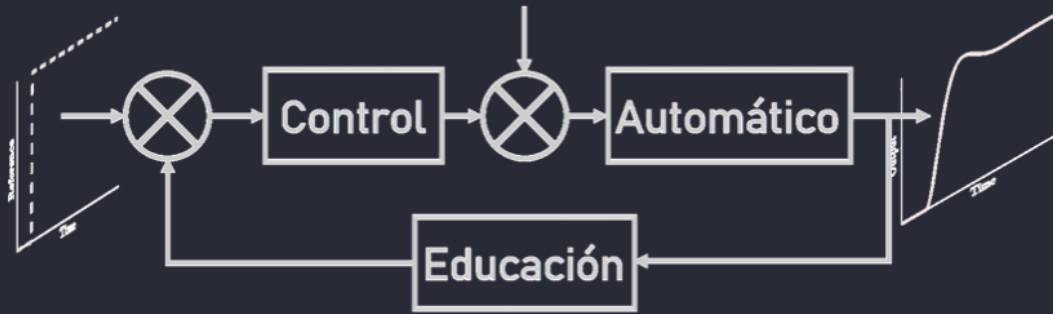




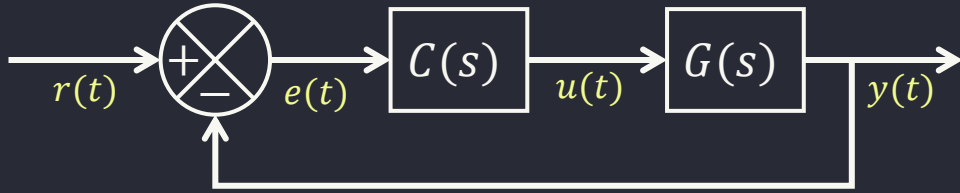
Margen de Ganancia y Margen de Fase

Sergio Andres Castaño Giraldo

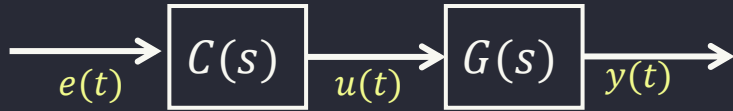
<https://controlautomaticoeducacion.com/>



Margen de Ganancia y Margen de Fase



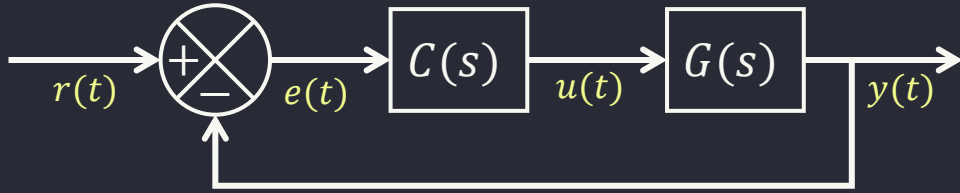
Lazo Abierto: $G(s)C(s)$



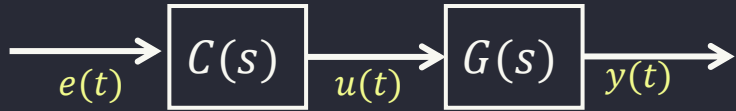
El Lazo Abierto DEBE ser ESTABLE
para el Análisis de BODE

Para poder saber si
la estabilidad
continúa al
momento de cerrar
el lazo

Margen de Ganancia y Margen de Fase

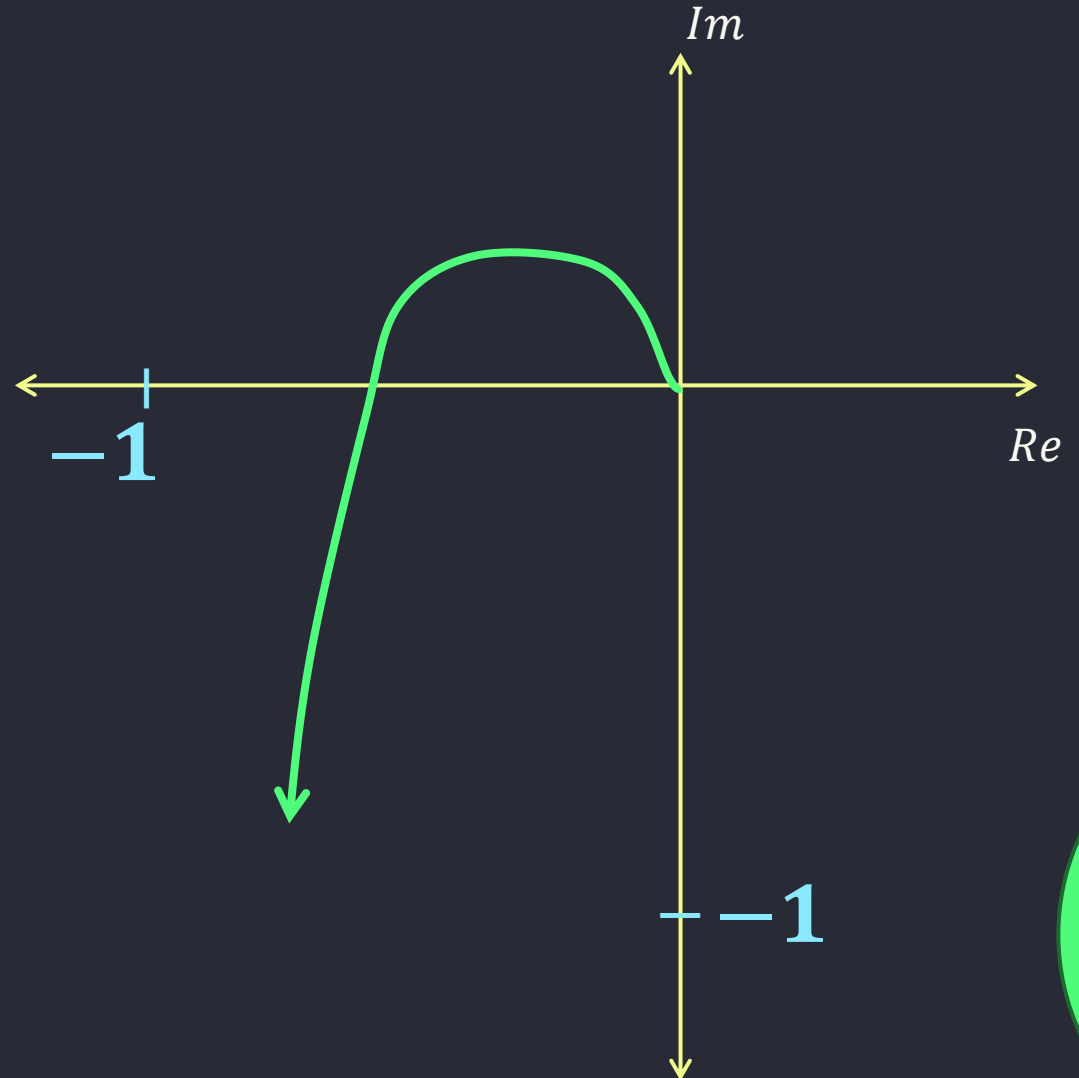


Lazo Abierto: $G(s)C(s)$

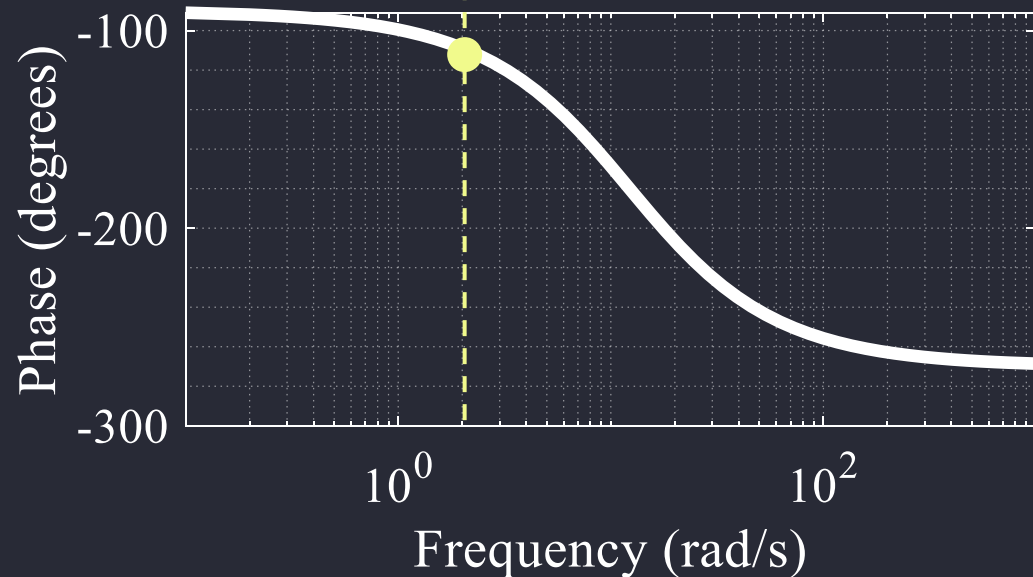
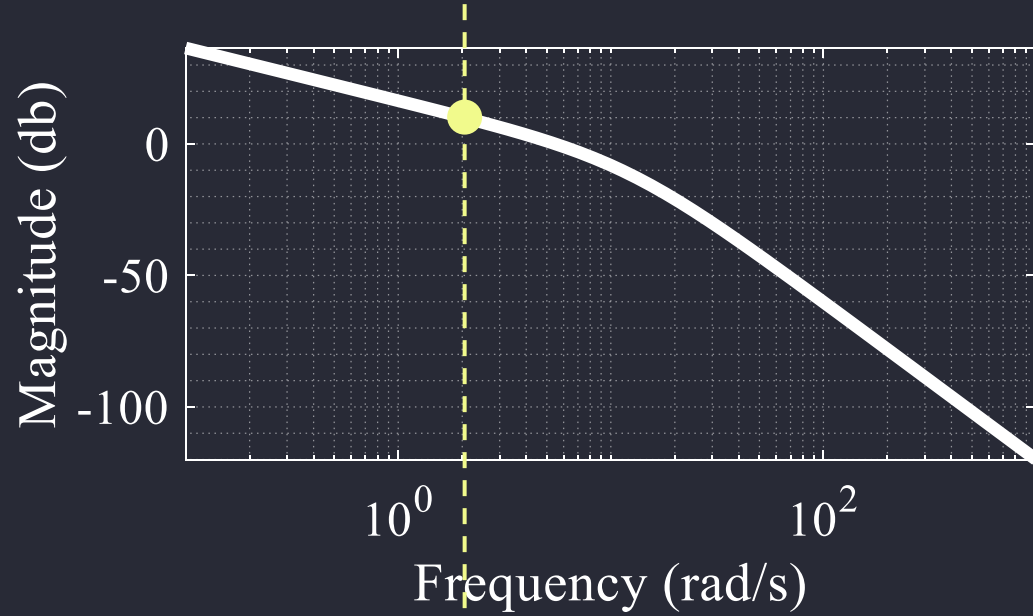


Conociendo el diagrama de bode de un sistema en lazo abierto, podemos determinar si el sistema en lazo cerrado será estable.

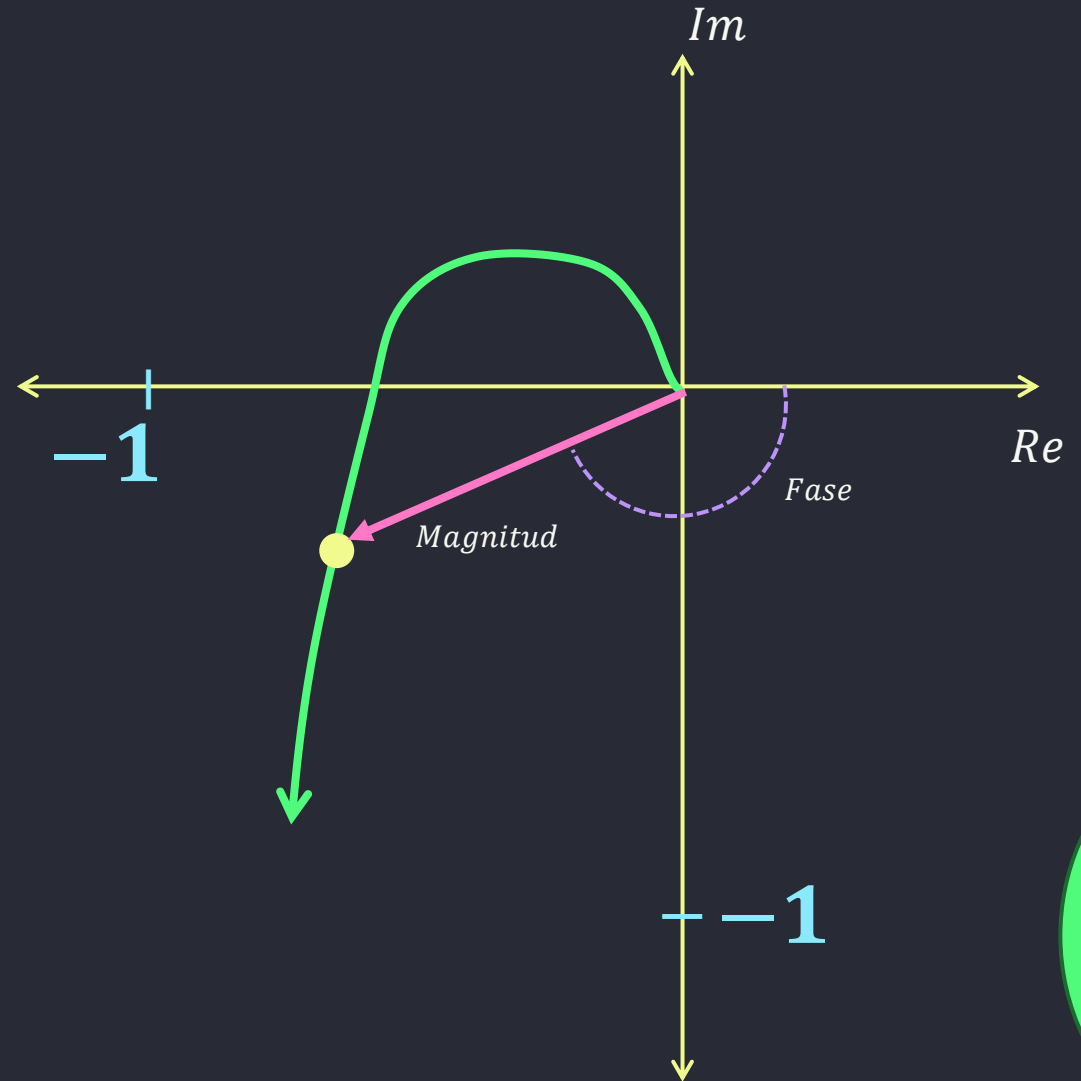
Diagrama de Nyquist:



Margen de Ganancia y Margen de Fase

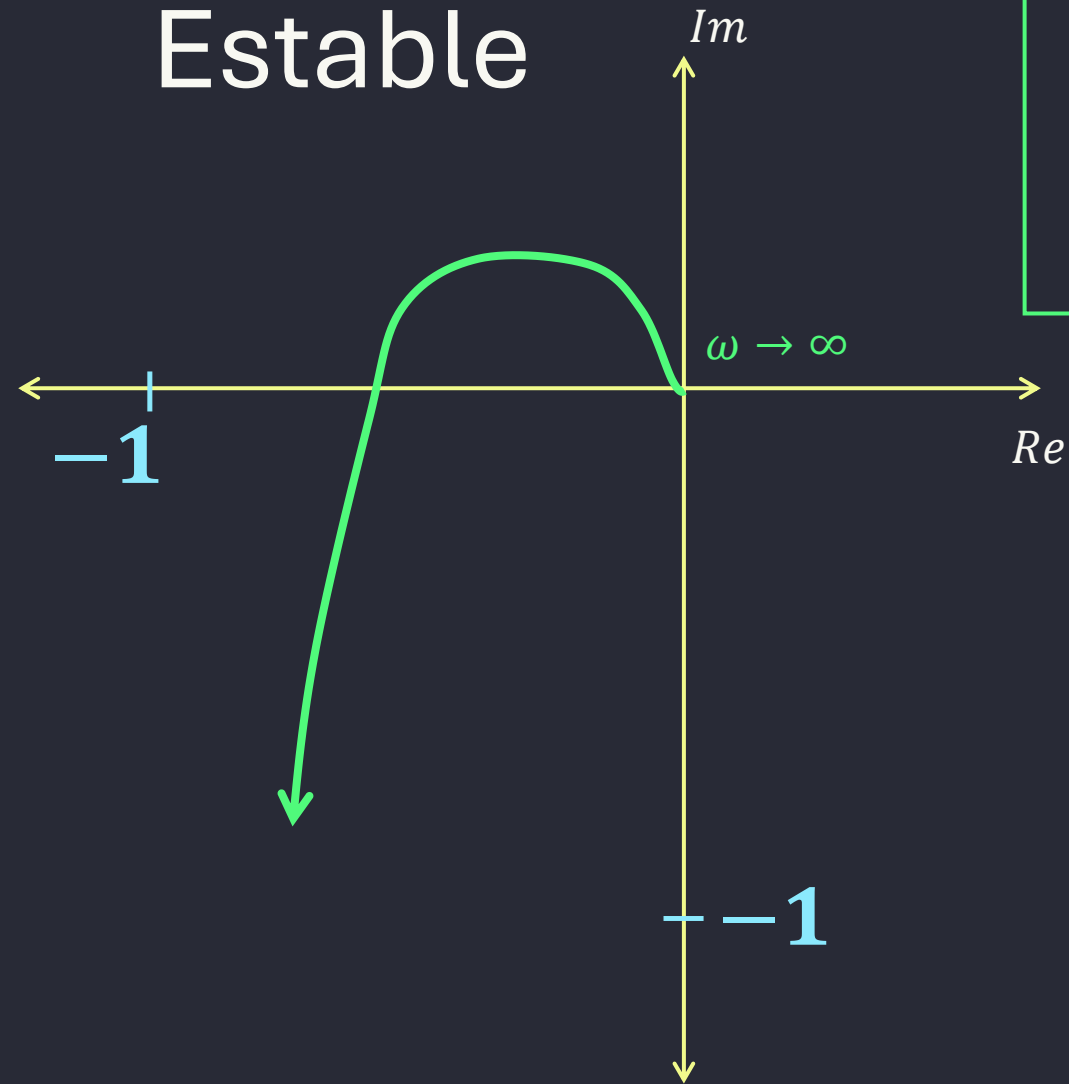


Lazo Abierto: $G(s)C(s)$



Margen de Ganancia y Margen de Fase

Estable



gm = Margen de Ganancia

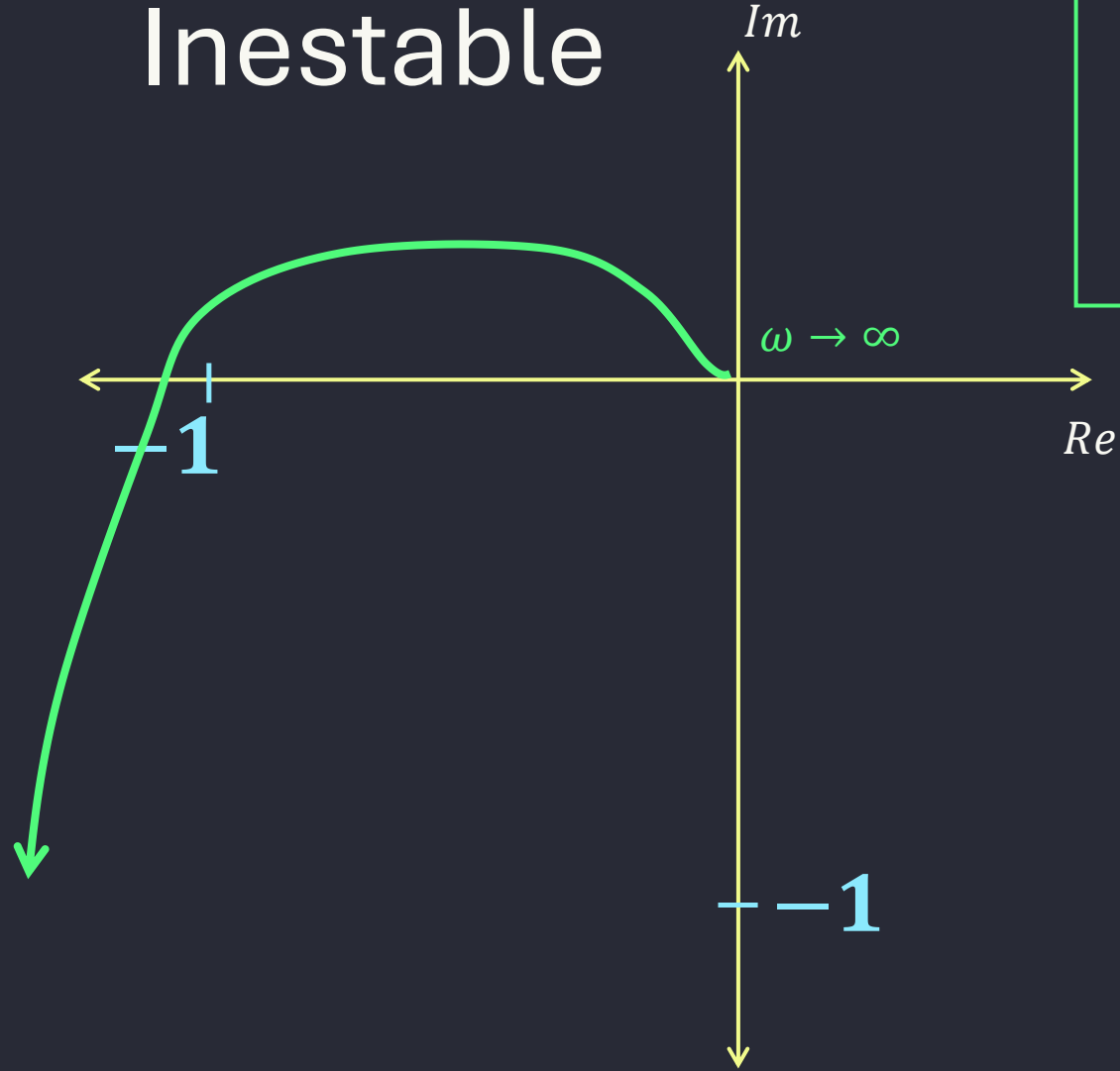
pm = Margen de Fase

W_{cg} = Mide gm

W_{cp} = Mide pm

Margen de Ganancia y Margen de Fase

Inestable



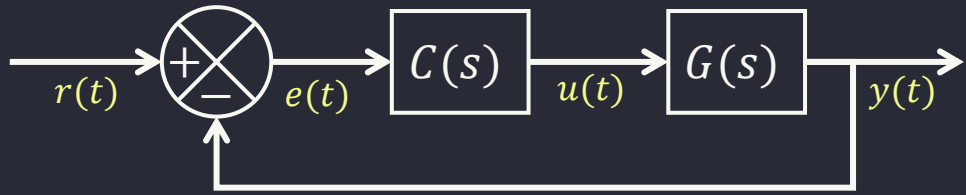
gm = Margen de Ganancia

pm = Margen de Fase

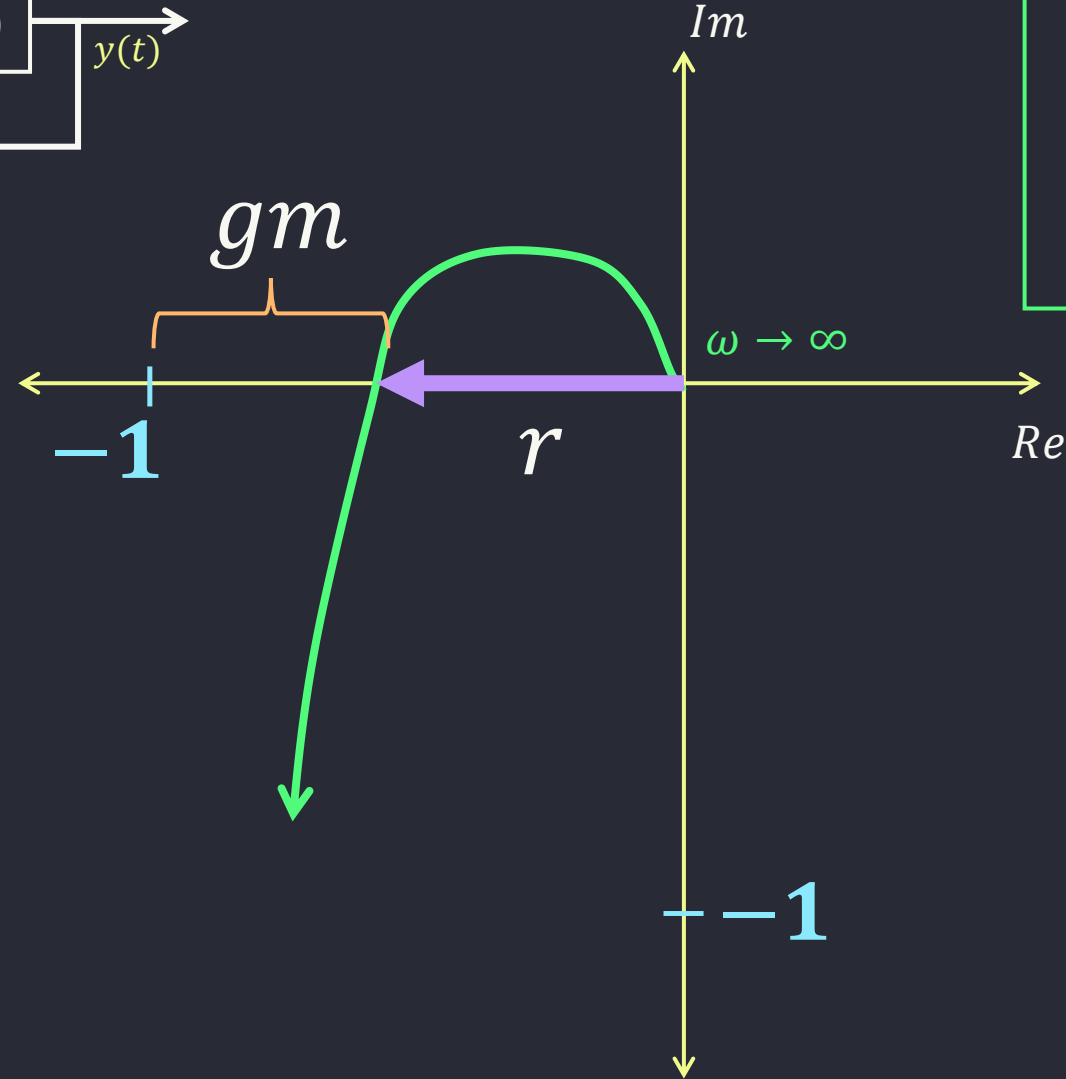
W_{cg} = Mide gm

W_{cp} = Mide pm

Margen de Ganancia y Margen de Fase



$$gm = \frac{1}{r}$$



gm = Margen de Ganancia

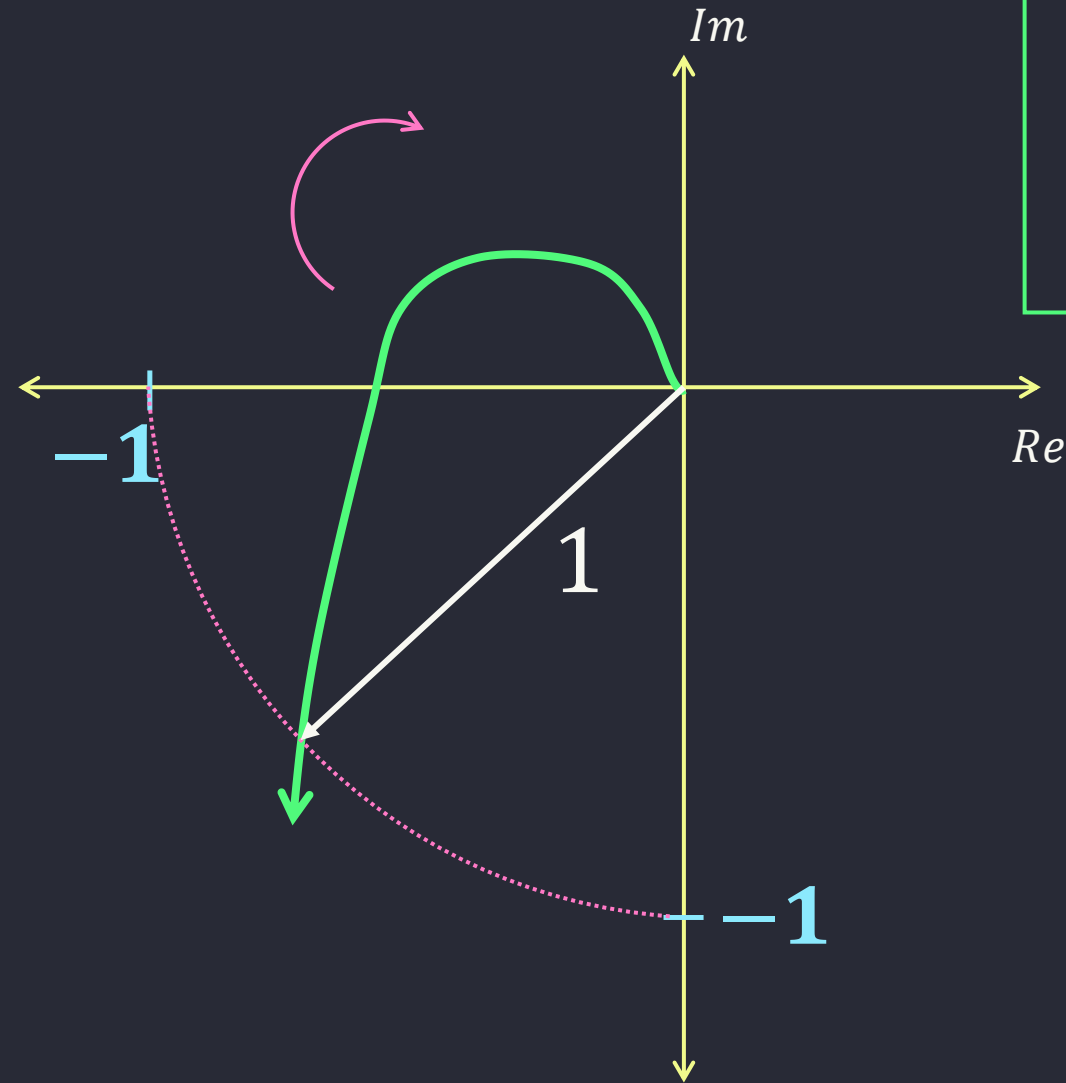
pm = Margen de Fase

W_{cg} = Mide gm

W_{cp} = Mide pm

Margen de Ganancia y Margen de Fase

$$gm = \frac{1}{r}$$



gm = Margen de Ganancia

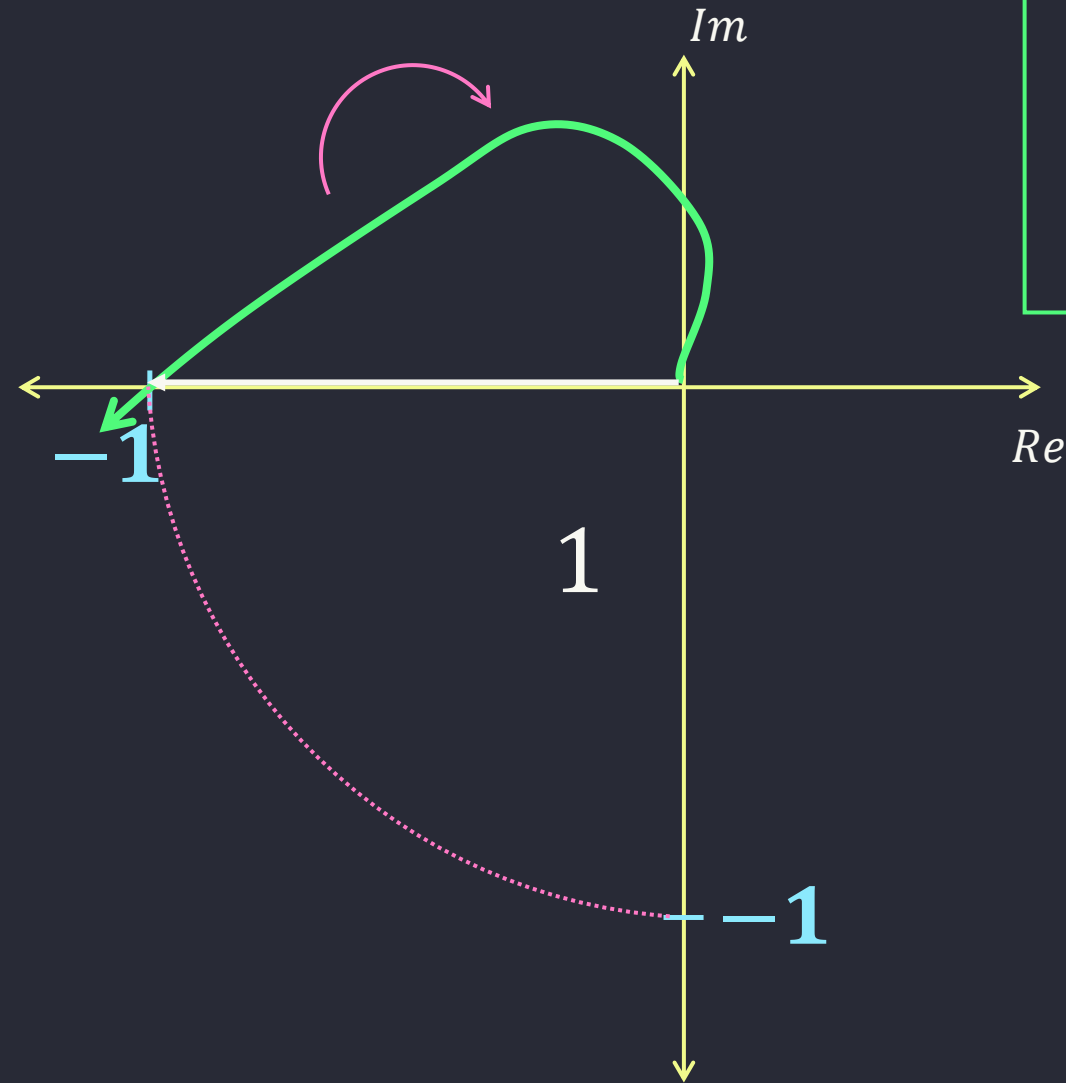
pm = Margen de Fase

W_{cg} = Mide gm

W_{cp} = Mide pm

Margen de Ganancia y Margen de Fase

$$gm = \frac{1}{r}$$



gm = Margen de Ganancia

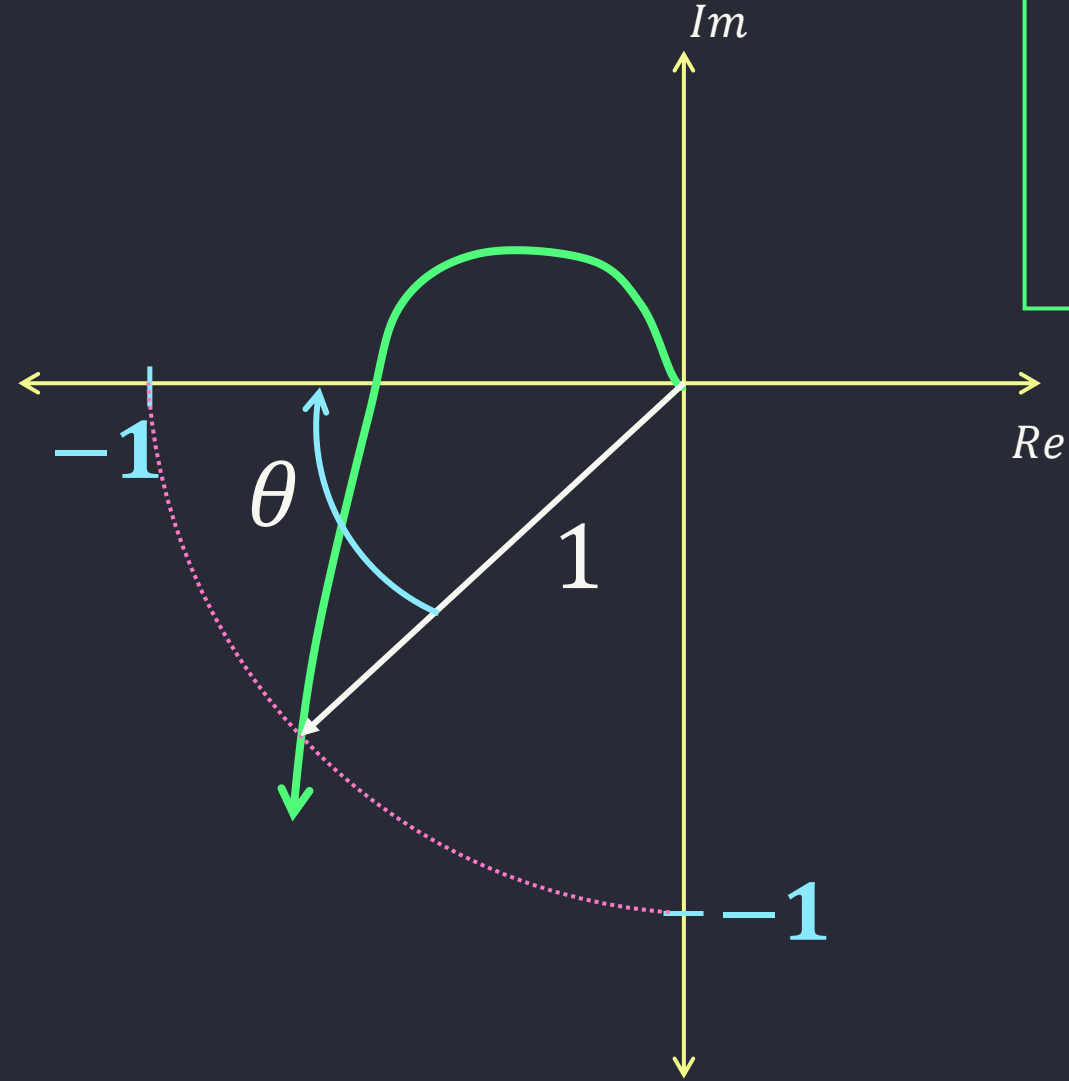
pm = Margen de Fase

W_{cg} = Mide gm

W_{cp} = Mide pm

Margen de Ganancia y Margen de Fase

$$gm = \frac{1}{r}$$
$$pm = \theta$$

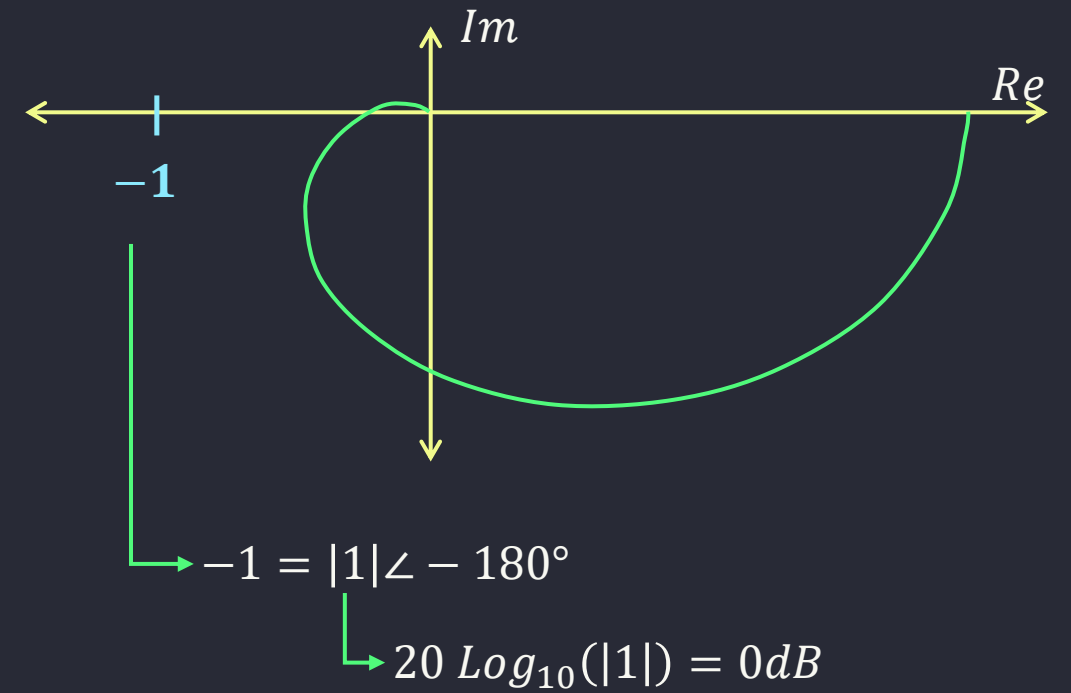
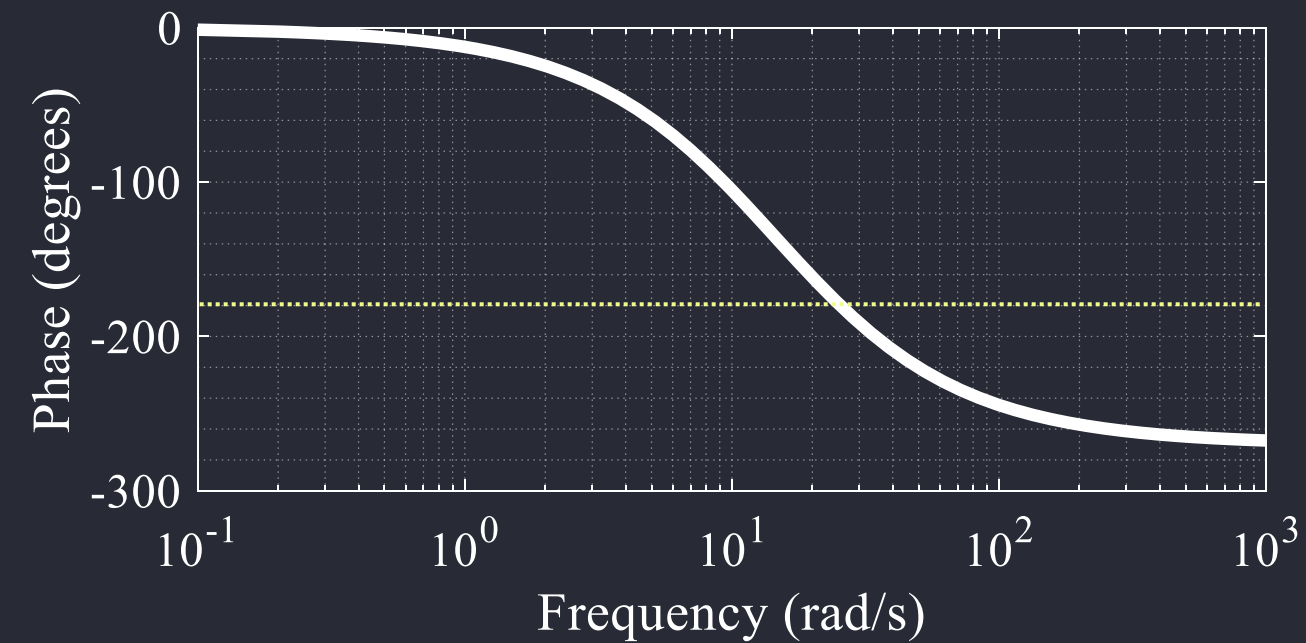
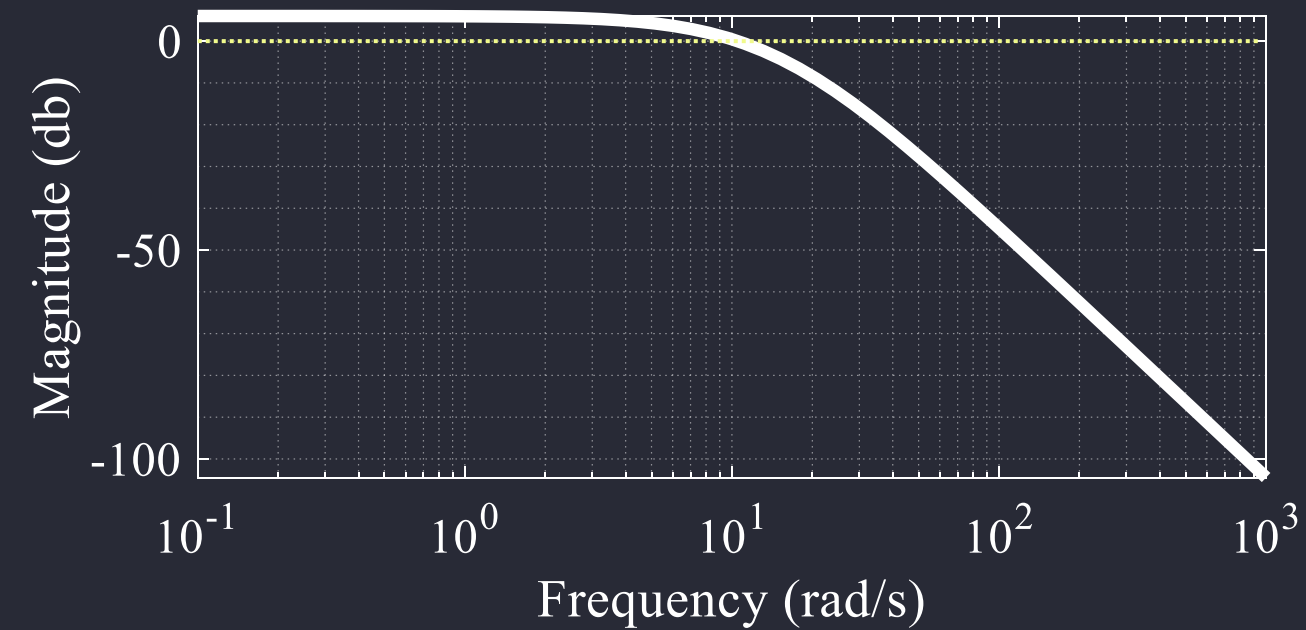


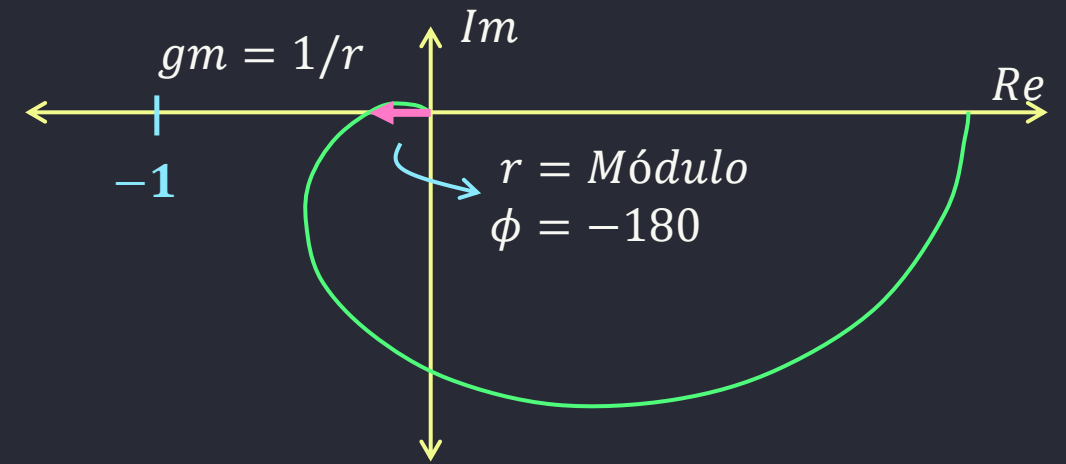
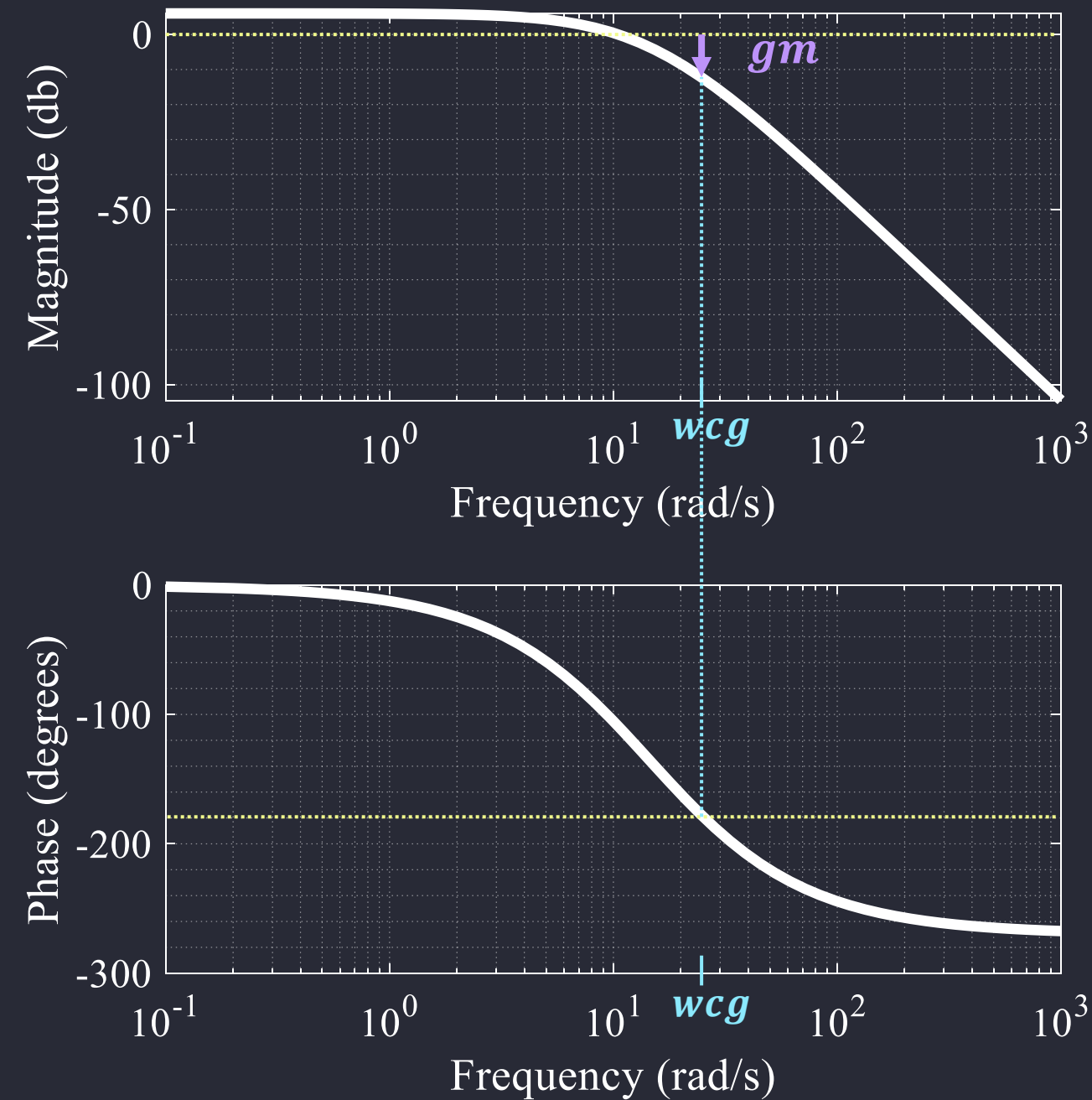
gm = Margen de Ganancia

pm = Margen de Fase

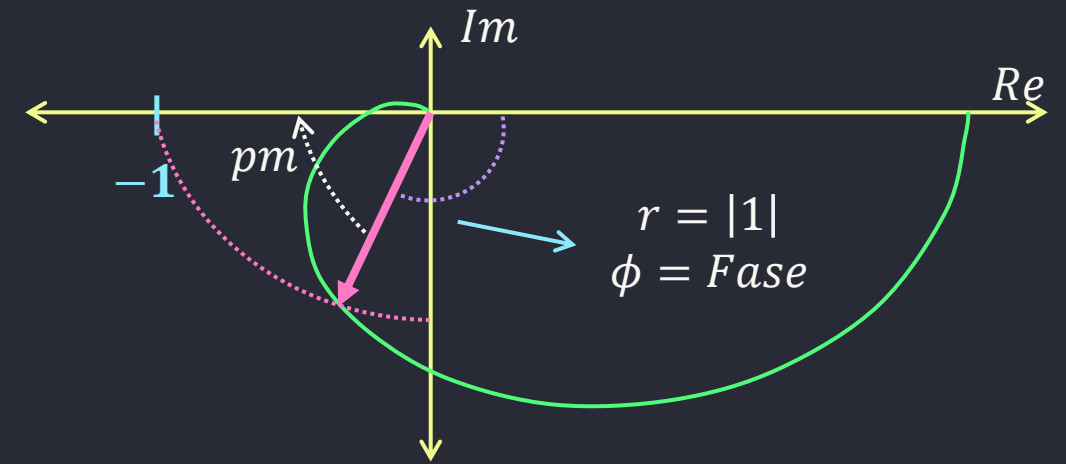
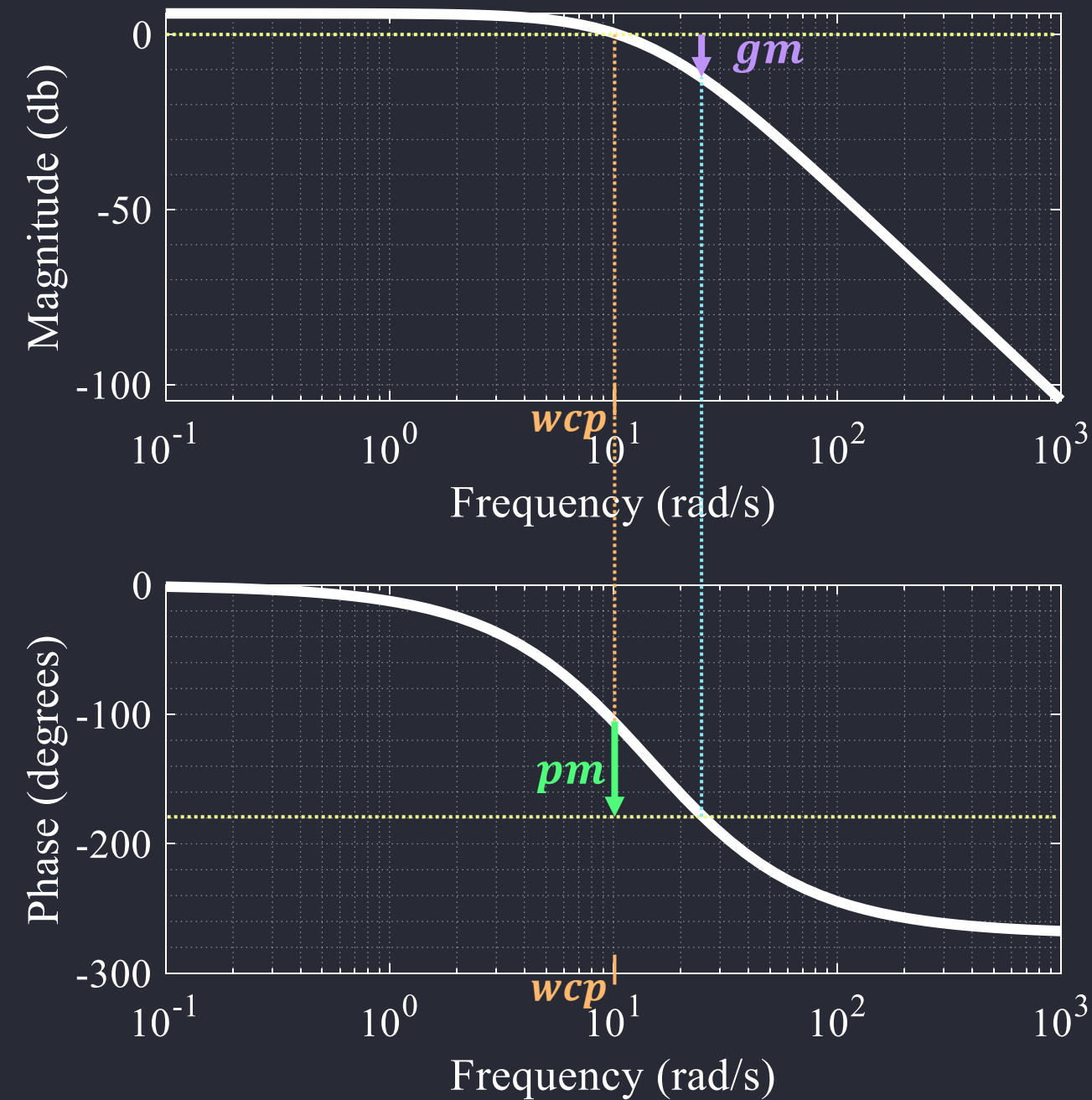
W_{cg} = Mide gm

W_{cp} = Mide pm

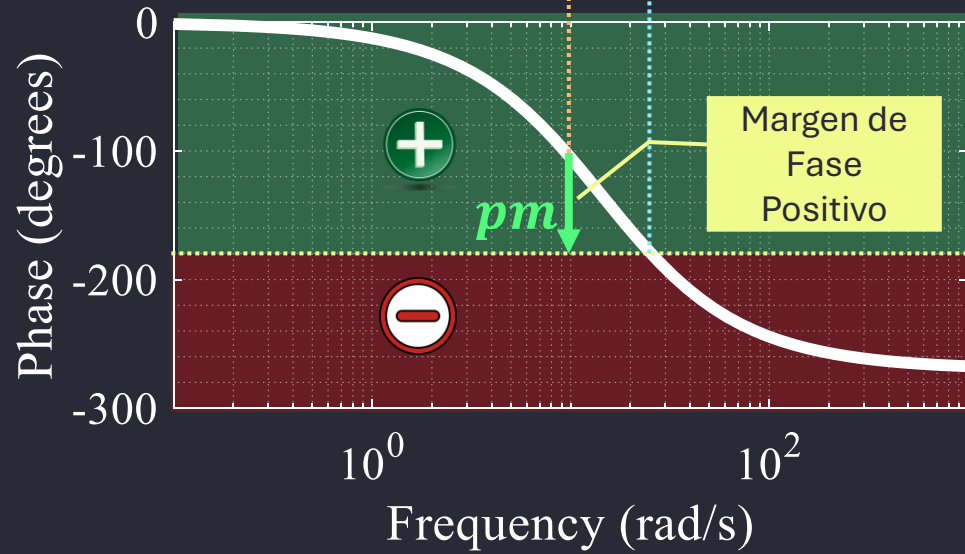
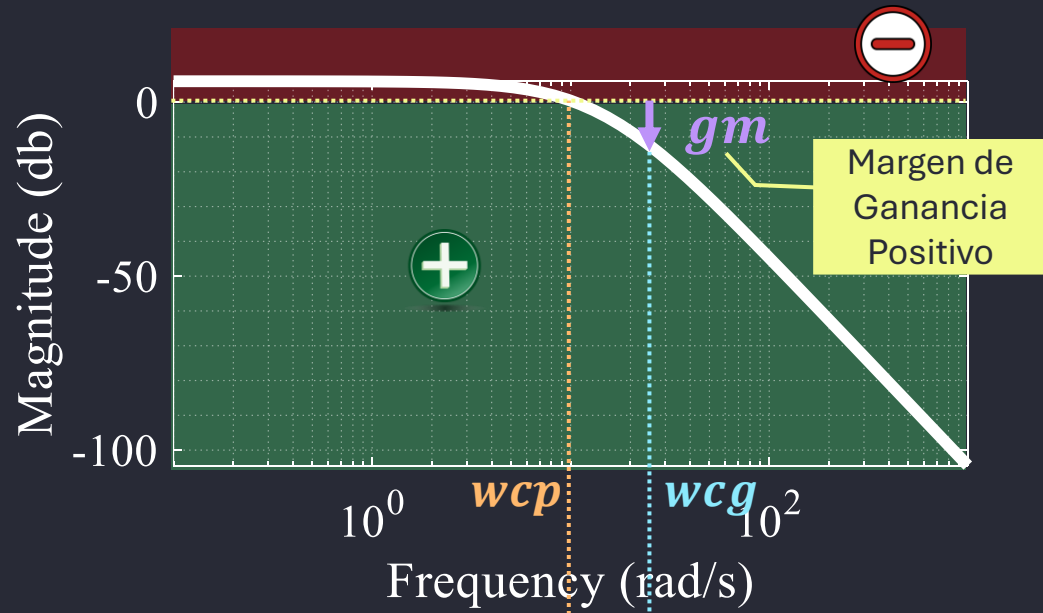




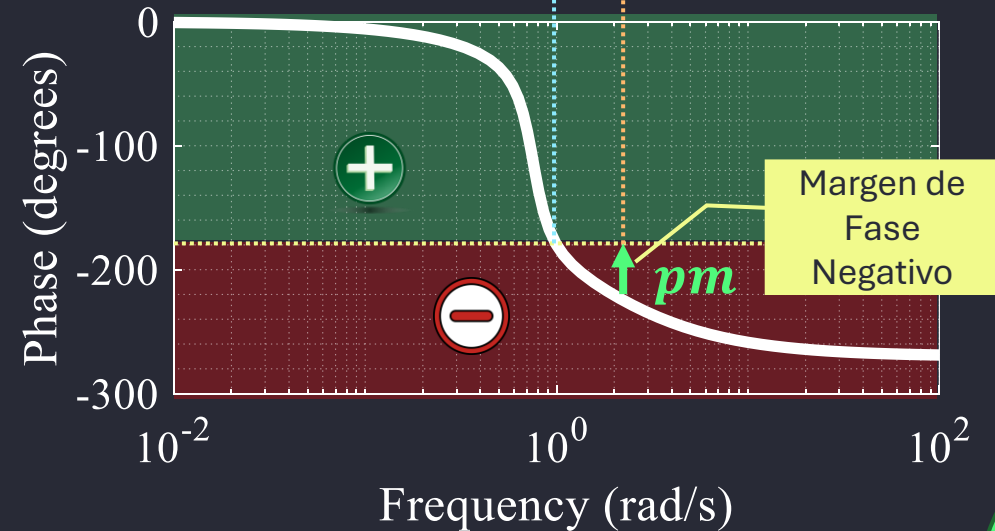
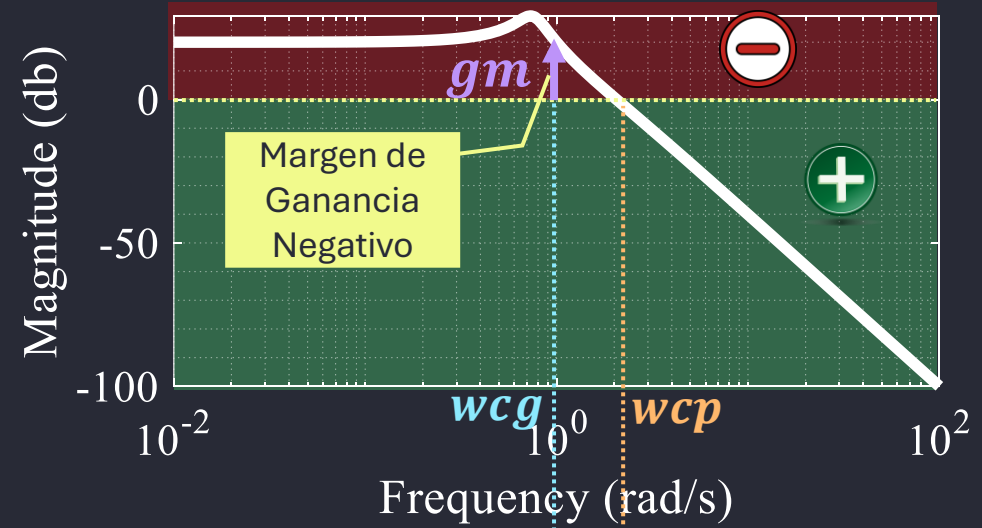
- **Margen de Ganancia (gm)** Se define como la magnitud del recíproco de la función de transferencia de lazo abierto, calculada a la **frecuencia de cruce de fase**.
- **Frecuencia de cruce de fase (w_{cg})**: Es la frecuencia a la cual el ángulo de fase de la función de transferencia de lazo abierto alcanza -180°



- **Margen de fase (pm):** Se define como la suma de 180° al ángulo θ^c , medido a la frecuencia de cruce de ganancia.
- **Frecuencia de Cruce de Ganancia (w_{cp}):** Se define como la frecuencia a la cual la magnitud de la función de transferencia de lazo abierto es igual a 1 es decir, 0 db

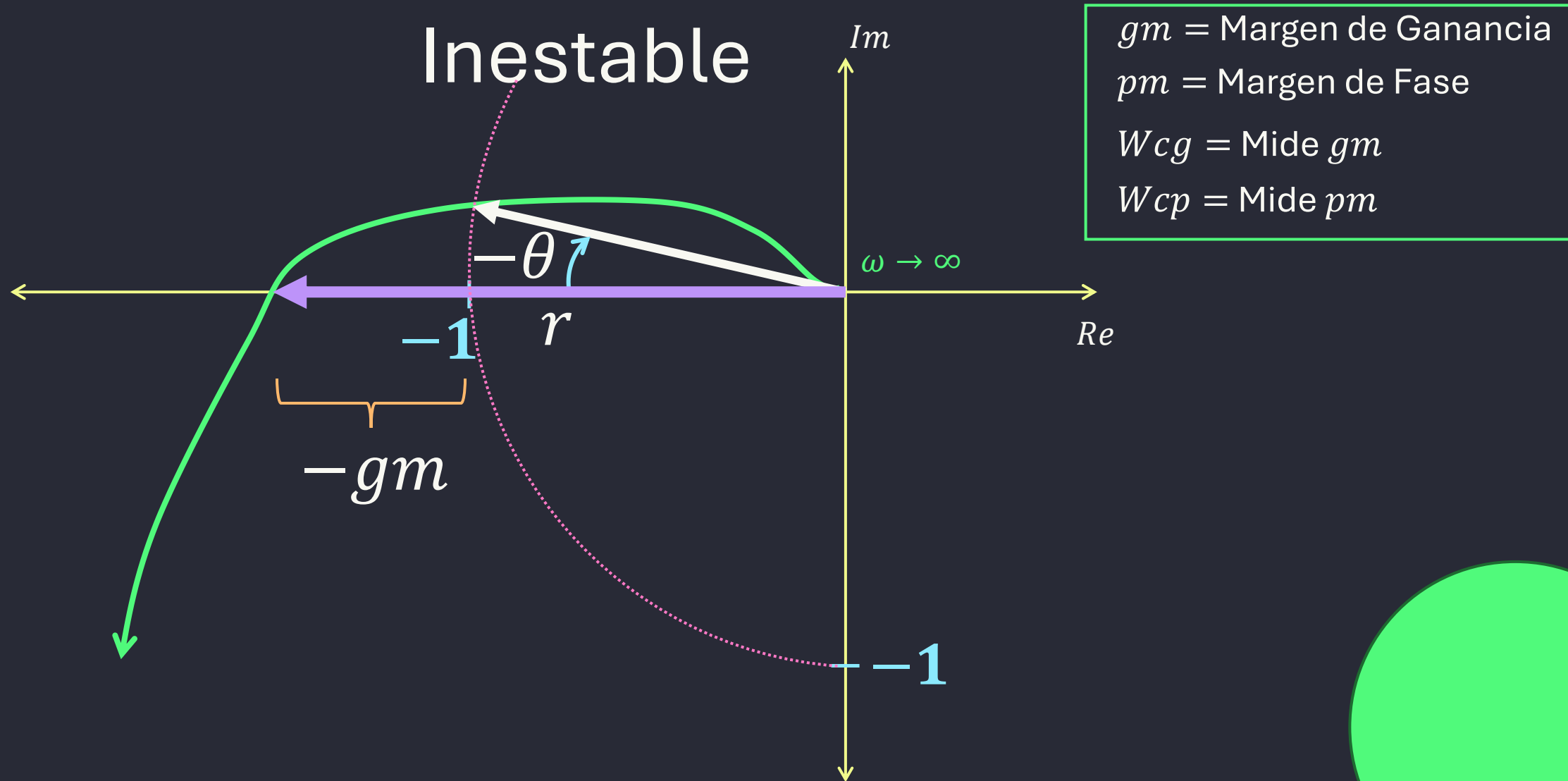


Estable
gm y pm deben ser positivos

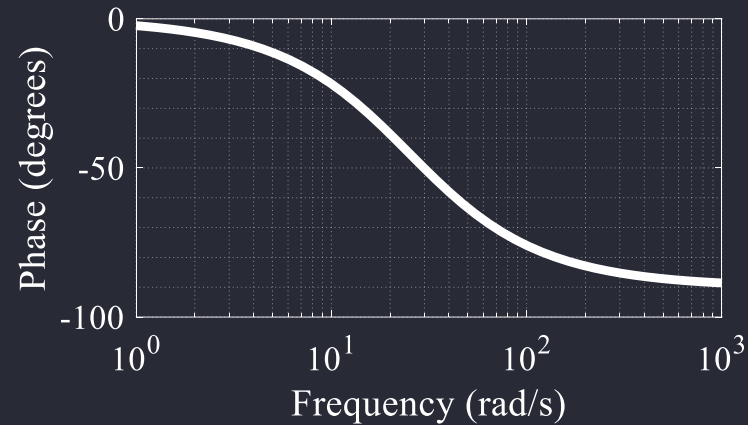
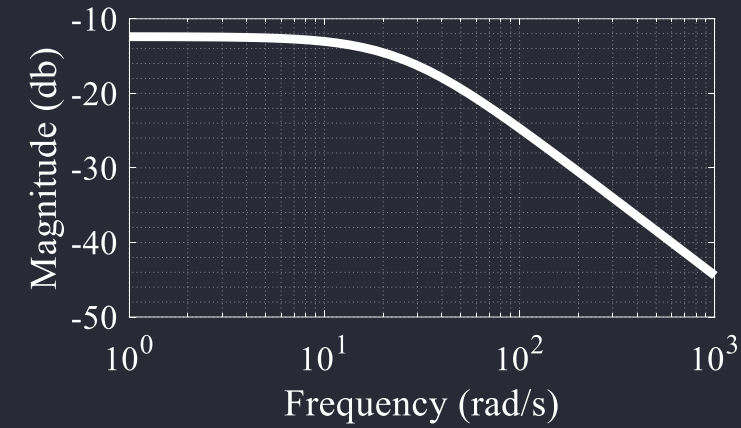


Inestable
Si al menos uno es negativo

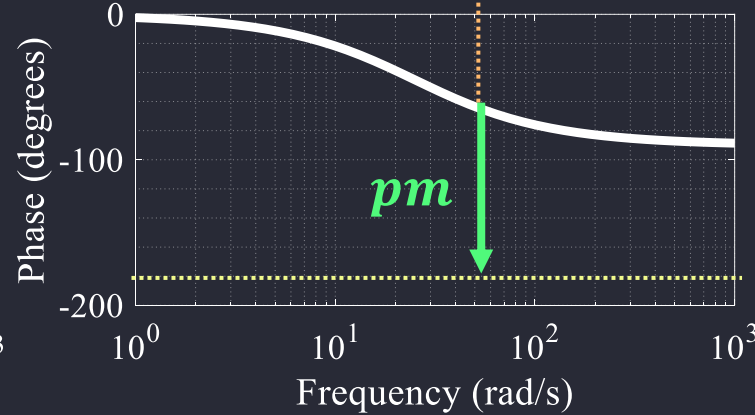
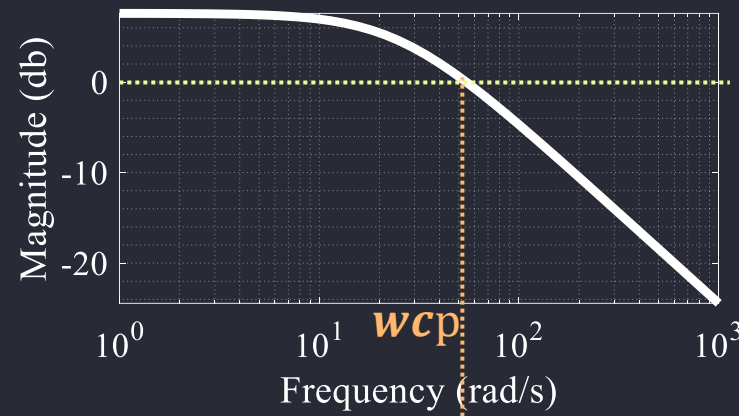
Margen de Ganancia y Margen de Fase



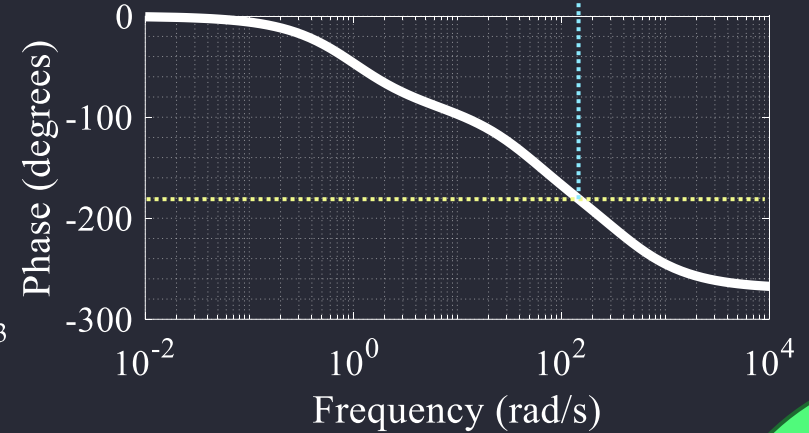
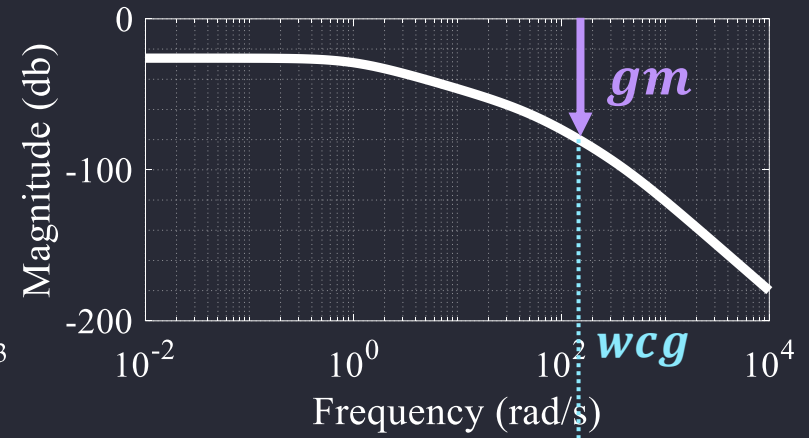
Casos Particulares Sistemas ESTABLES



$$gm = \infty ; pm = \infty$$



$$gm = \infty ; pm = 115^\circ$$



$$gm = 80 ; pm = \infty$$

Recomendaciones Generales:

Margen de Ganancia (GM):

- 1. Recomendación:** Un margen de ganancia mayor que **6 dB** es generalmente recomendado.
- 2. Rango Típico:** Mayor que **6 a 12 dB**.
- 3. Motivo:** Un margen de ganancia de 6 dB significa que puedes duplicar la ganancia del sistema sin que se vuelva inestable. Esto proporciona una buena robustez frente a incrementos en la ganancia del controlador. Un margen mayor, como 12 dB, indica una robustez aún mayor, pero podría hacer que el sistema sea menos eficiente en términos de control.

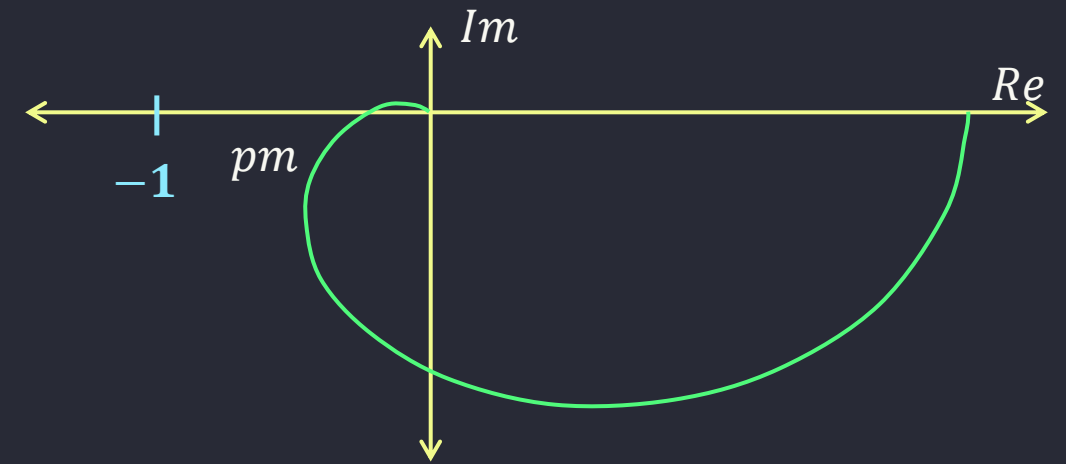
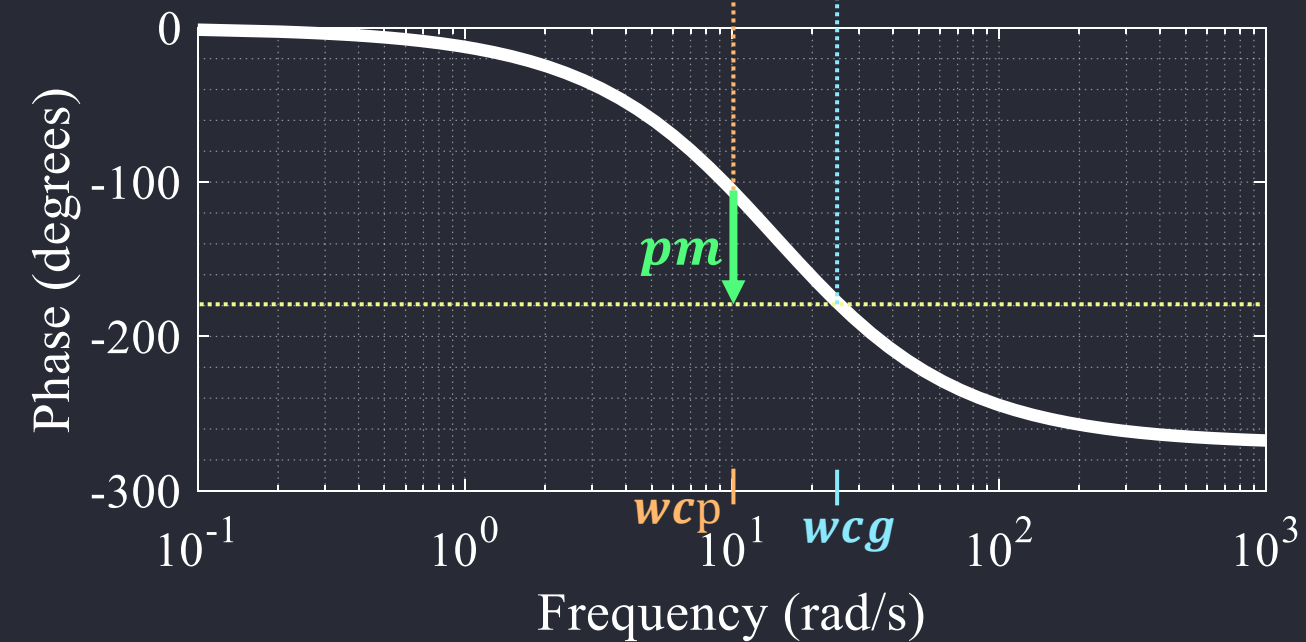
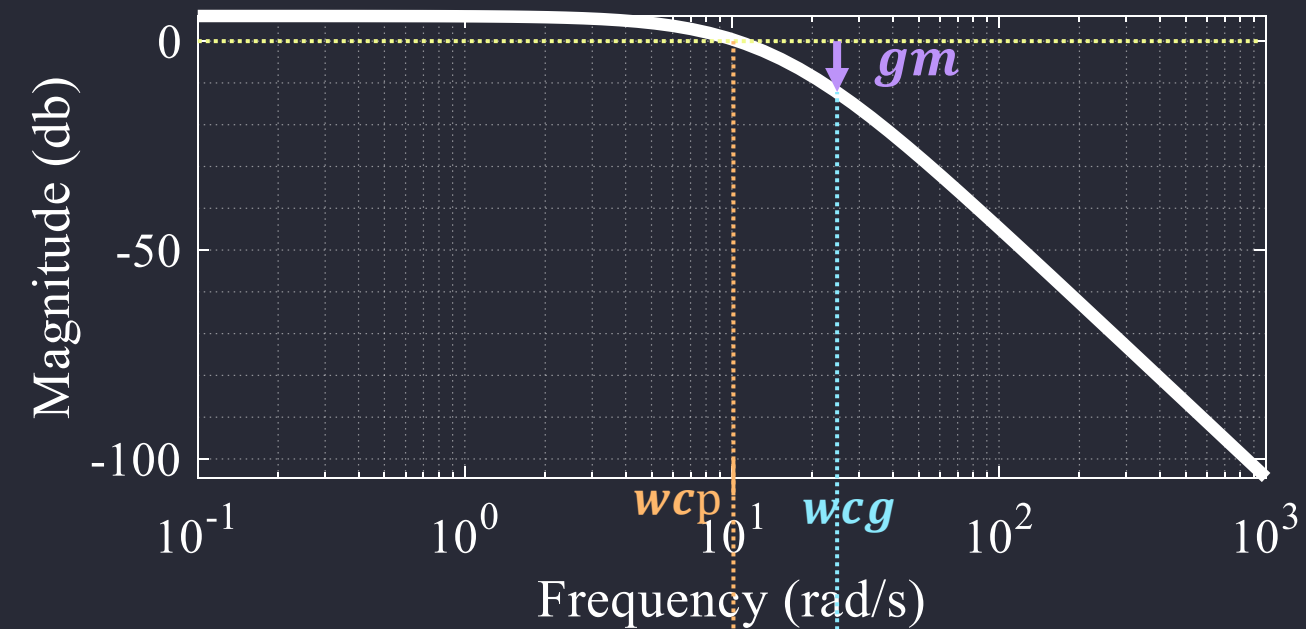


Recomendaciones Generales:

Margen de Fase (PM):

- 1. Recomendación:** Un margen de fase de al menos **45 grados** es generalmente recomendado.
- 2. Rango Típico:** De **30 a 60 grados**.
- 3. Motivo:** Un margen de fase de 45 grados proporciona un buen equilibrio entre estabilidad y capacidad de respuesta. Un margen menor de 30 grados puede indicar que el sistema es propenso a oscilaciones y puede volverse inestable con perturbaciones o variaciones en el sistema. Un margen mayor de 60 grados es muy estable, pero podría hacer que el sistema sea demasiado lento en su respuesta.





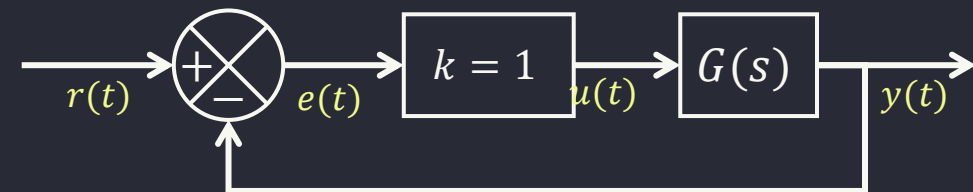
$$gm = 4.375 (Abs)$$

$$gm = 20 \log(4.375) = 12.8 dB$$

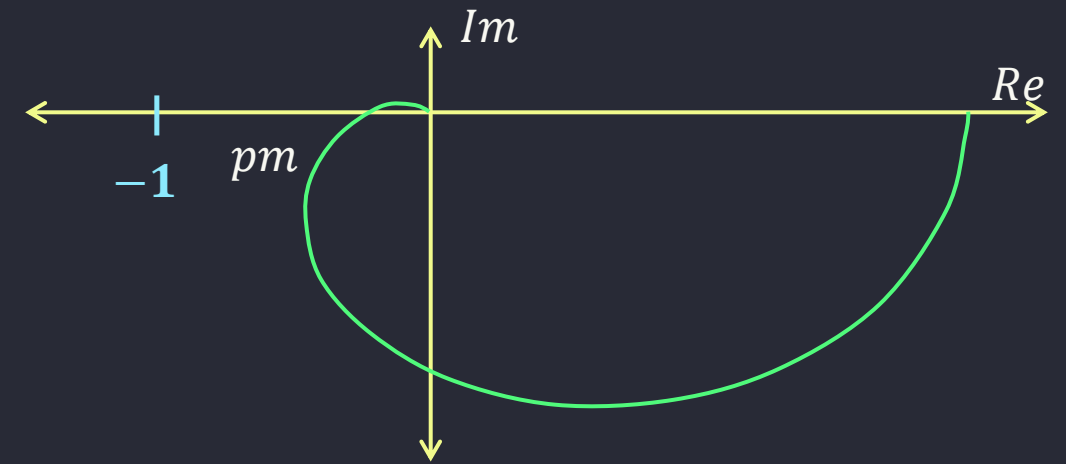
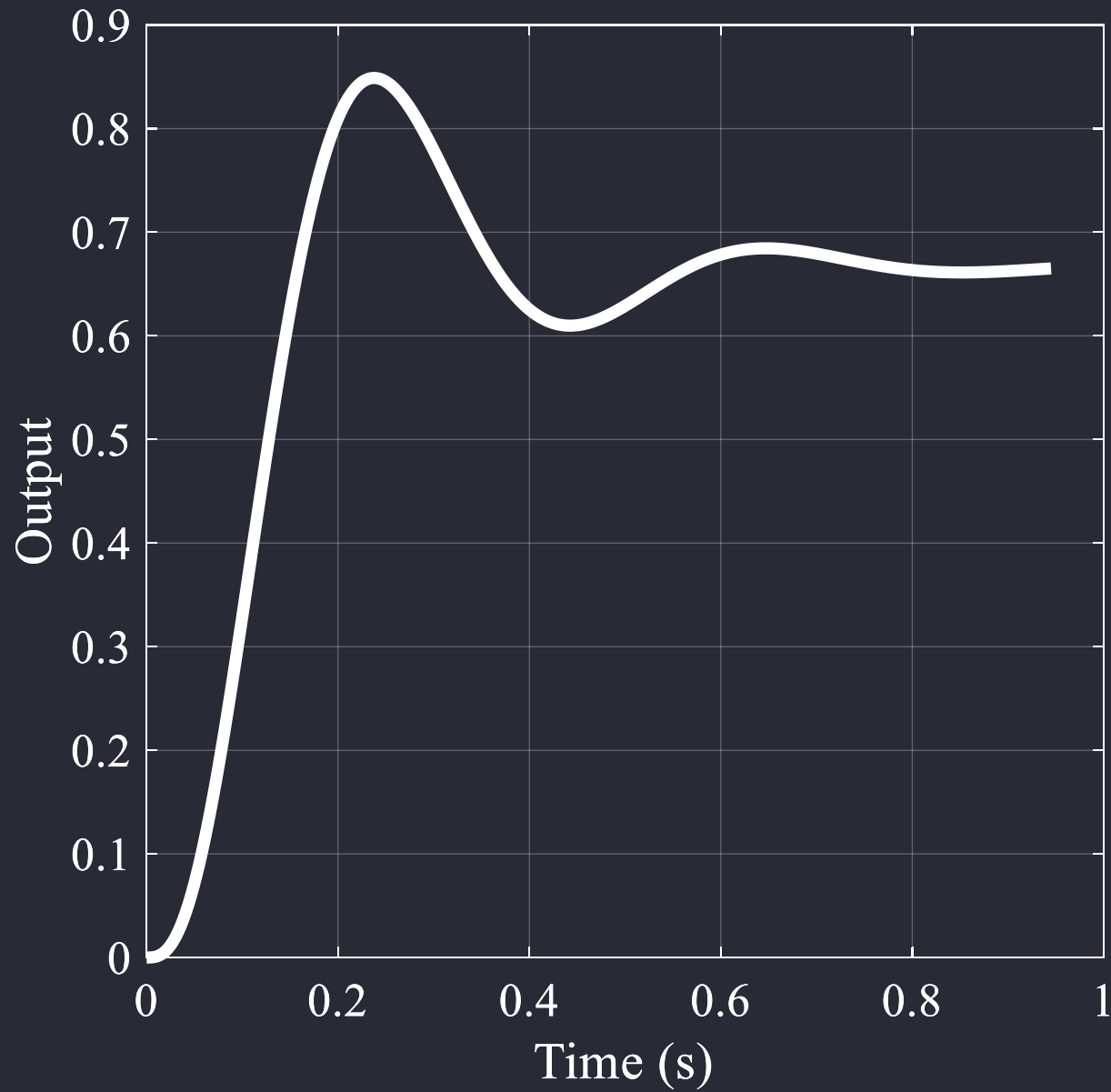
$$pm = 70.9^\circ$$

Estable

$$\text{Aumento Ganancia: } k_{gm} = 1(4.375) = 4.375$$



$$0 < k < 4.375$$



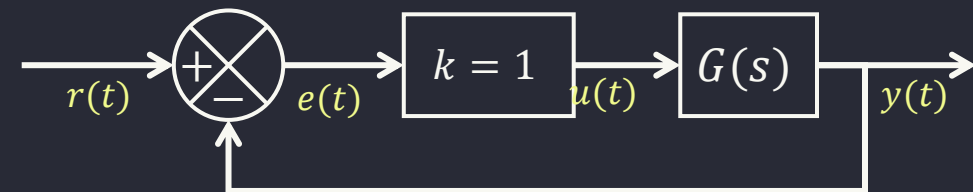
$$gm = 4.375 \text{ (Abs)}$$

$$gm = 20 \log(4.375) = 12.8 \text{ dB}$$

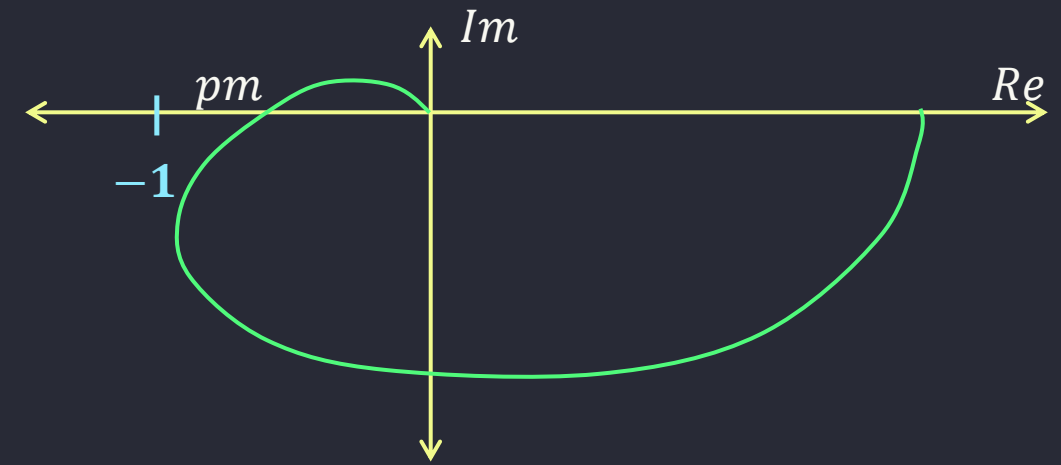
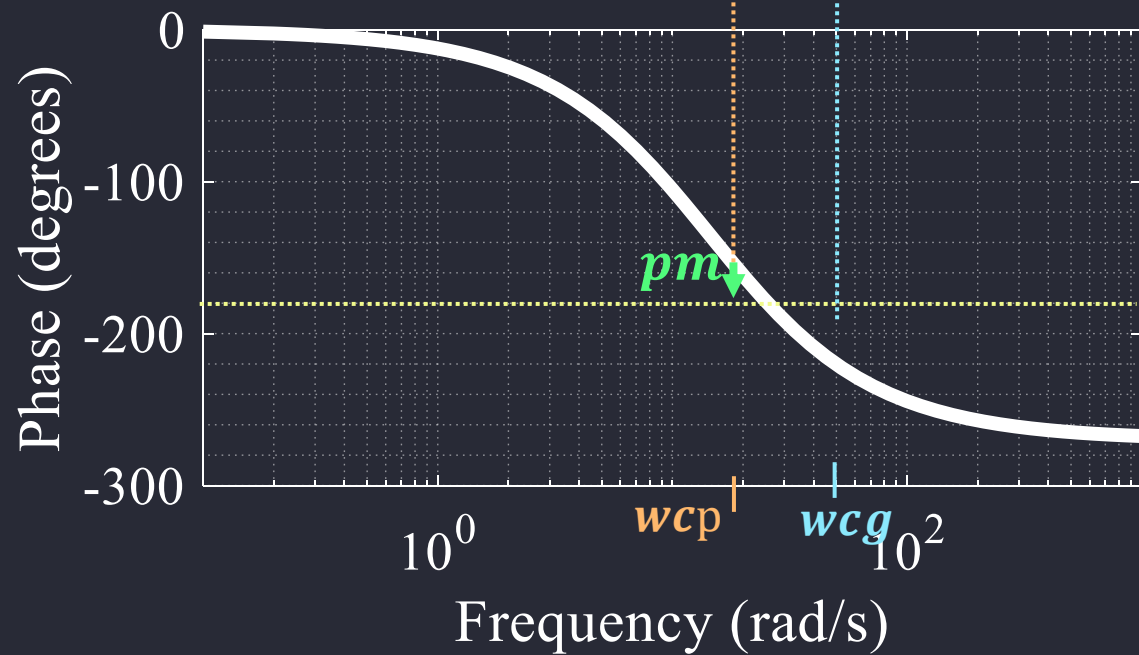
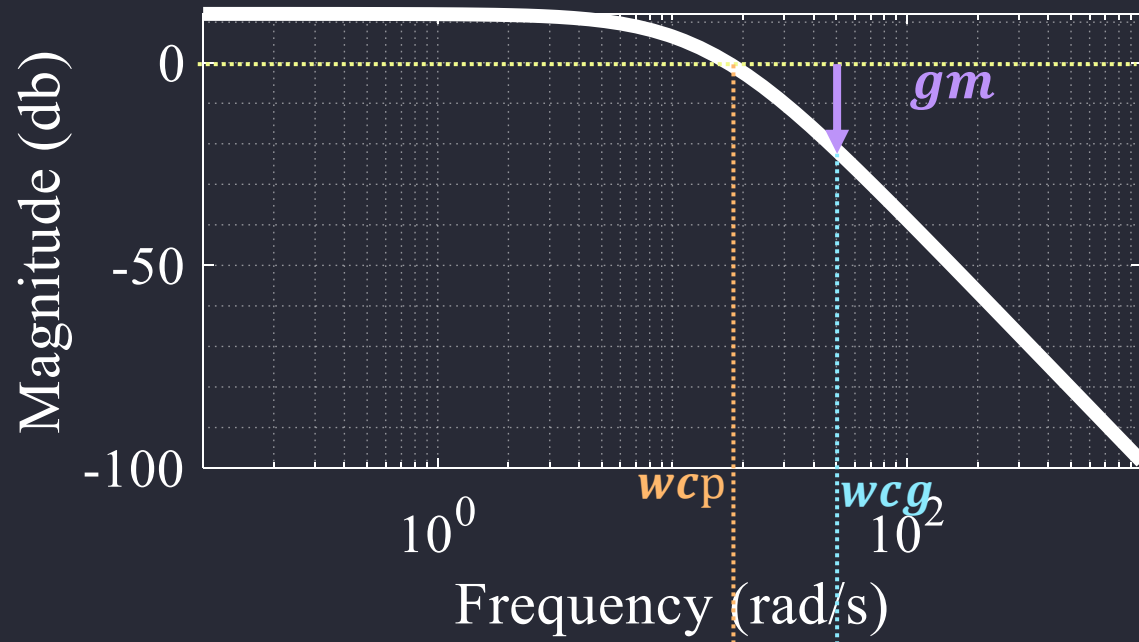
$$pm = 70.9^\circ$$

Estable

$$\text{Aumento Ganancia: } k_{gm} = 1(4.375) = 4.375$$



$$0 < k < 4.375$$



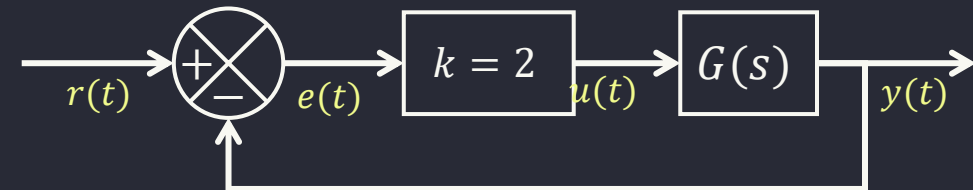
$$gm = 2.1875 \text{ (Abs)}$$

$$gm = 20 \text{Log}(4.375) = 6.8 \text{ dB}$$

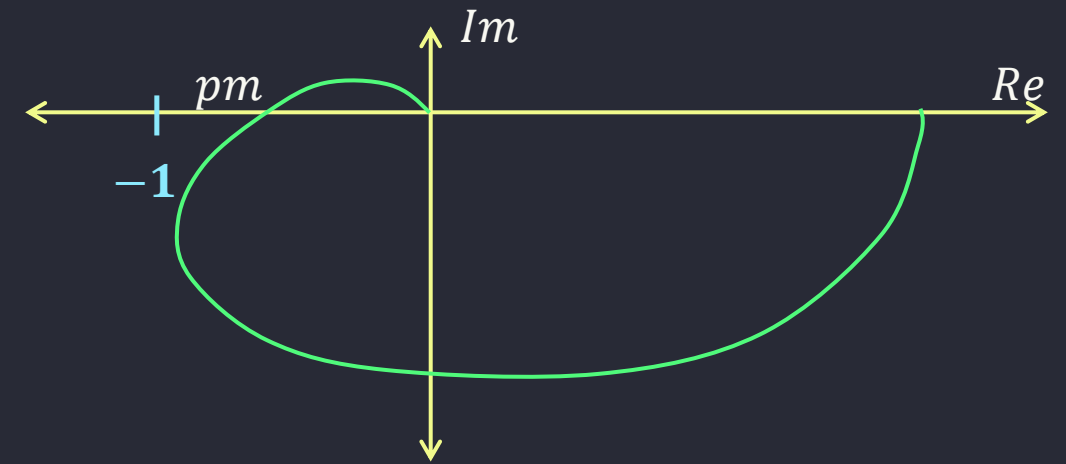
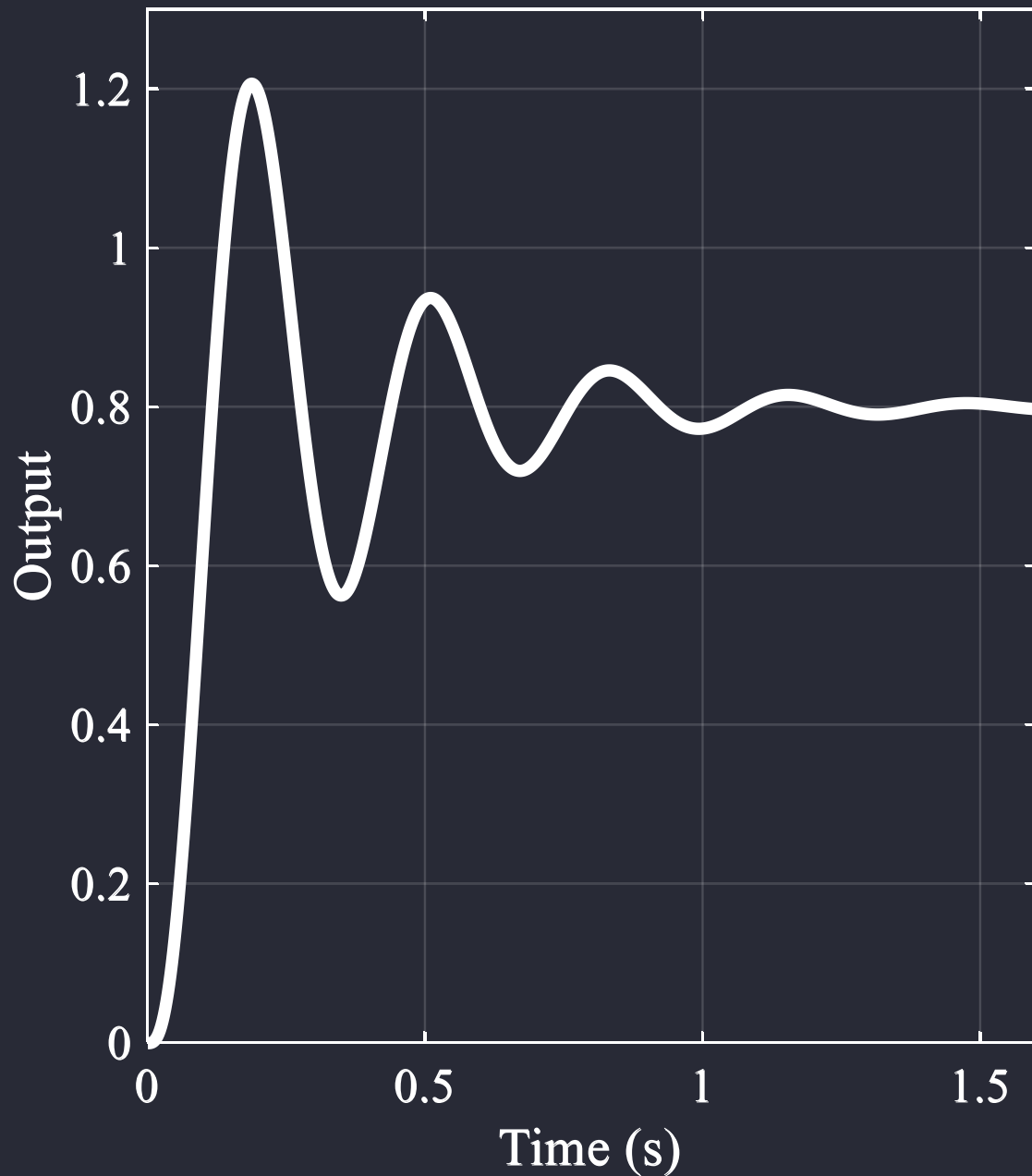
$$pm = 30.5^\circ$$

Estable

$$\text{Aumento Ganancia: } k_{gm} = 2(2.1875) = 4.375$$



$$0 < k < 4.375$$



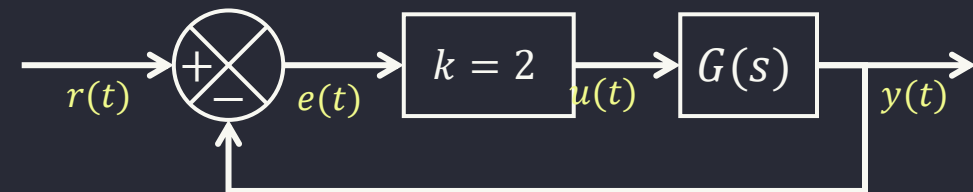
$$gm = 2.1875 \text{ (Abs)}$$

$$gm = 20\text{Log}(4.375) = 6.8\text{dB}$$

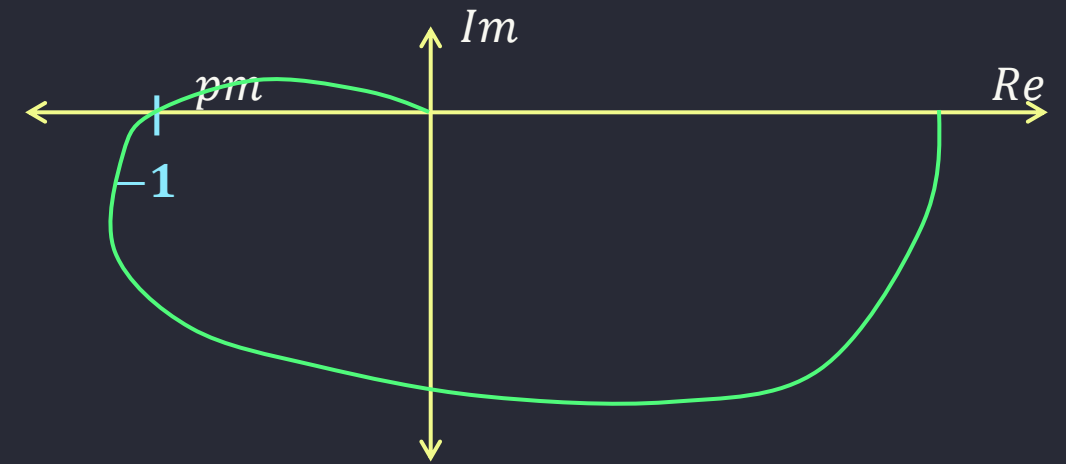
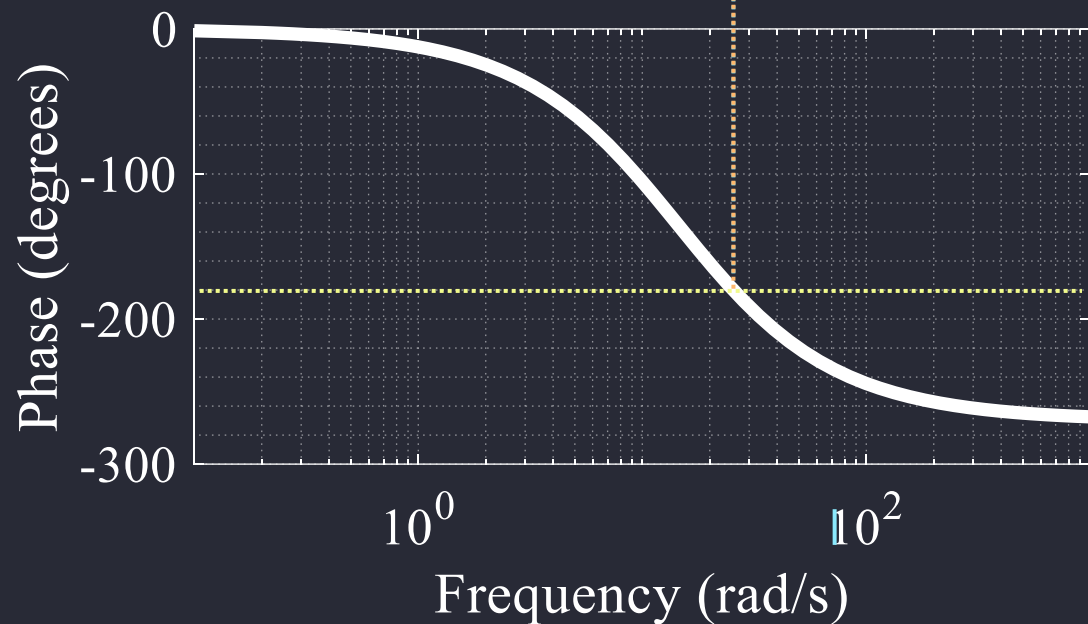
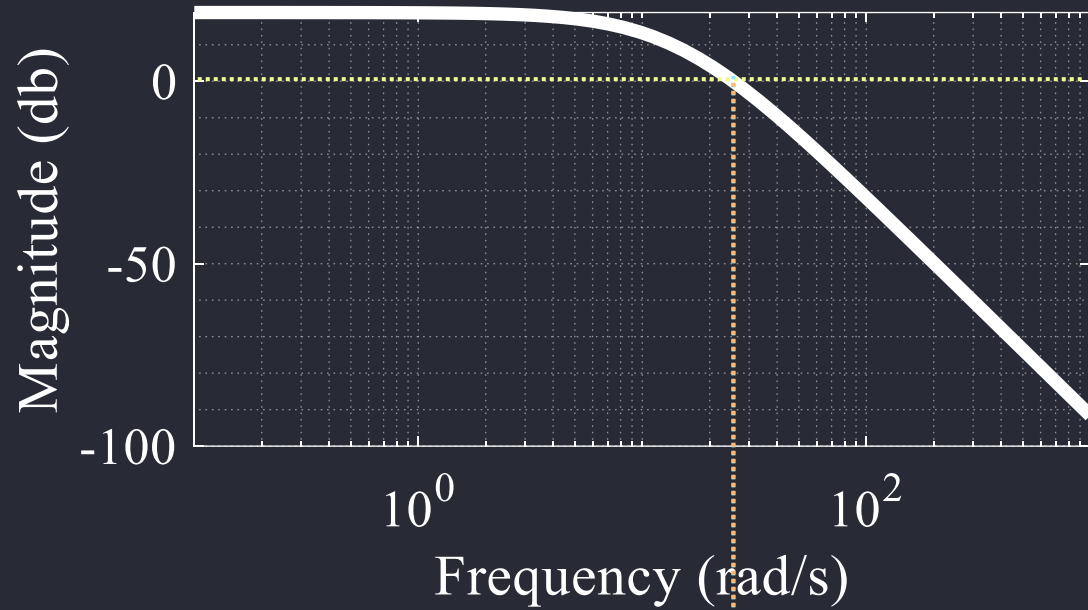
$$pm = 30.5^\circ$$

} Estable

$$\text{Aumento Ganancia: } k_{gm} = 2(2.1875) = 4.375$$



$$0 < k < 4.375$$



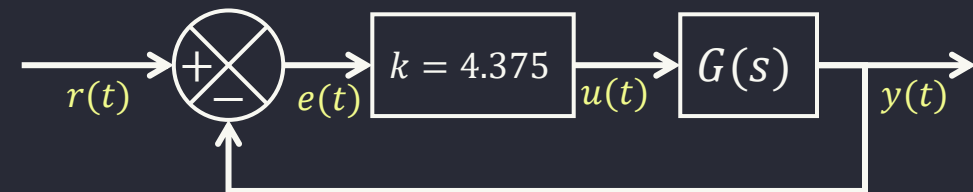
$$gm = 1 (Abs)$$

$$gm = 20 \log(1) = 0 dB$$

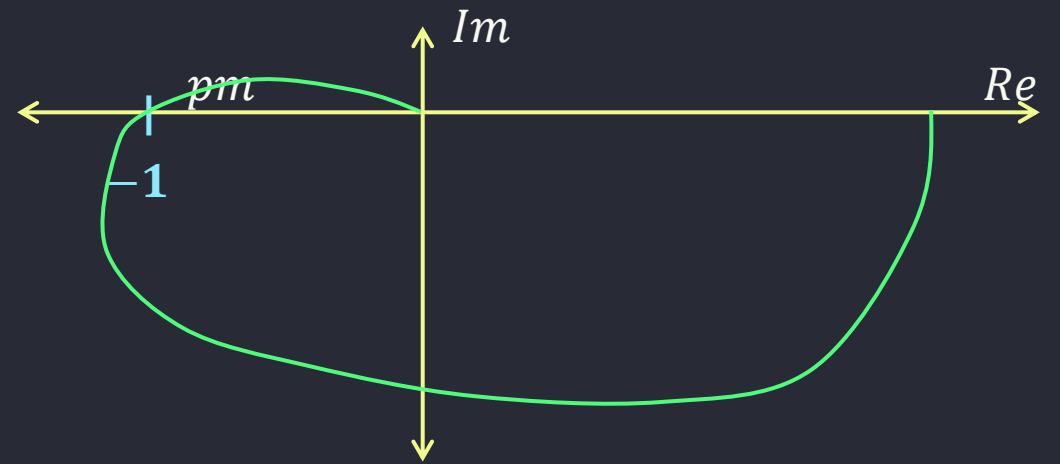
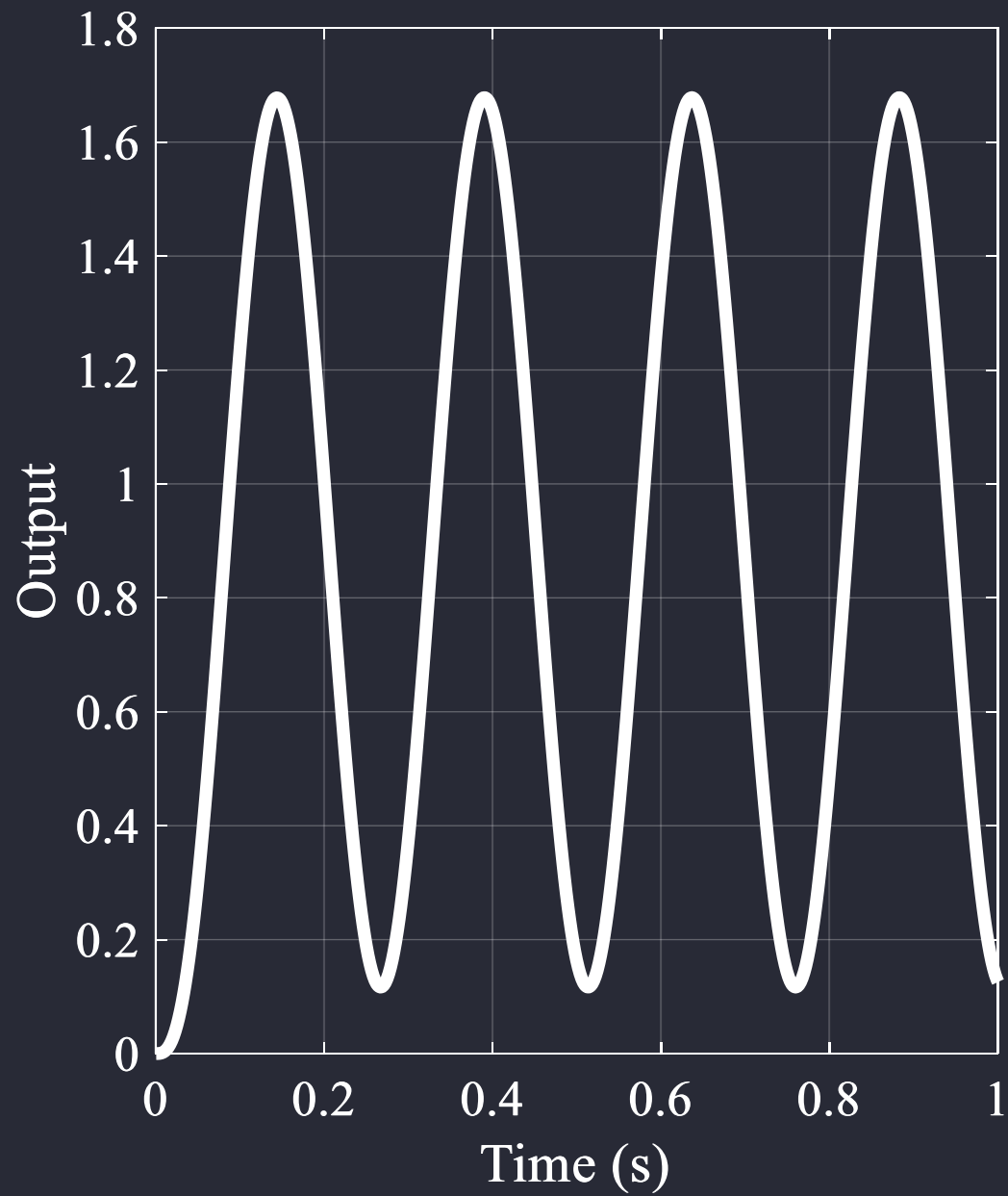
$$pm = 0^\circ$$

Marginalmente Estable

$$\text{Aumento Ganancia: } k_{gm} = 4.375(1) = 4.375$$



$$0 < k < 4.375$$



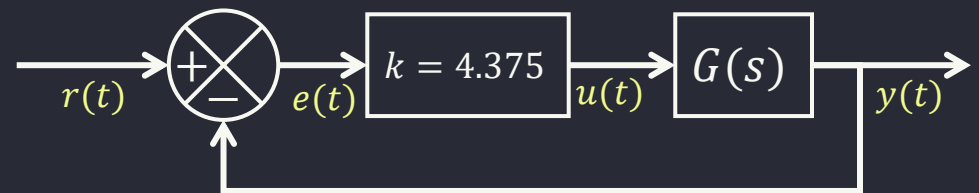
$$gm = 1 (Abs)$$

$$gm = 20Log(1) = 0dB$$

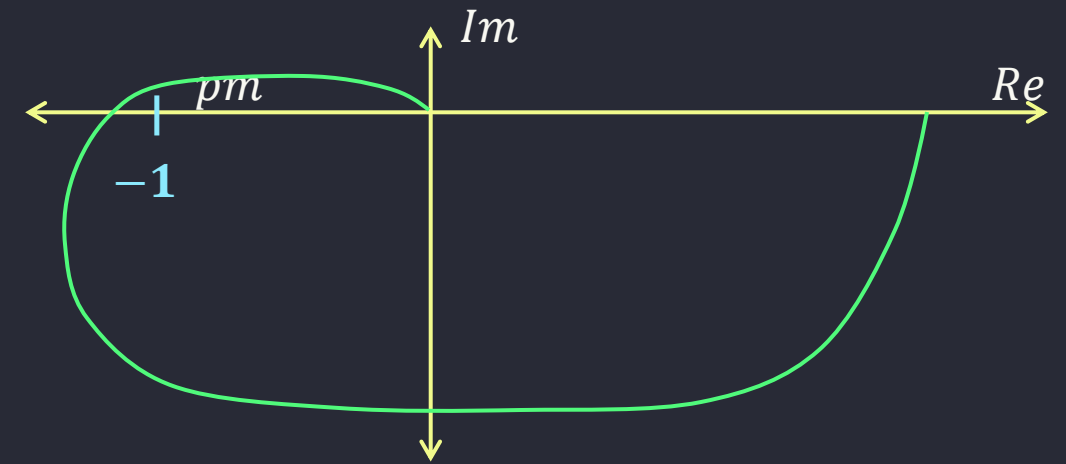
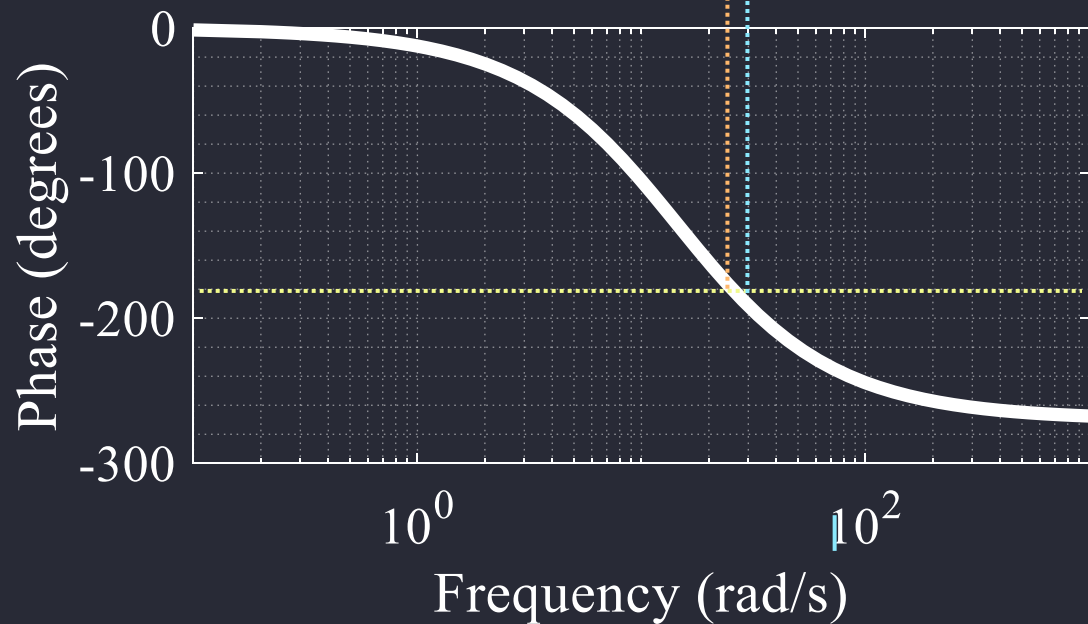
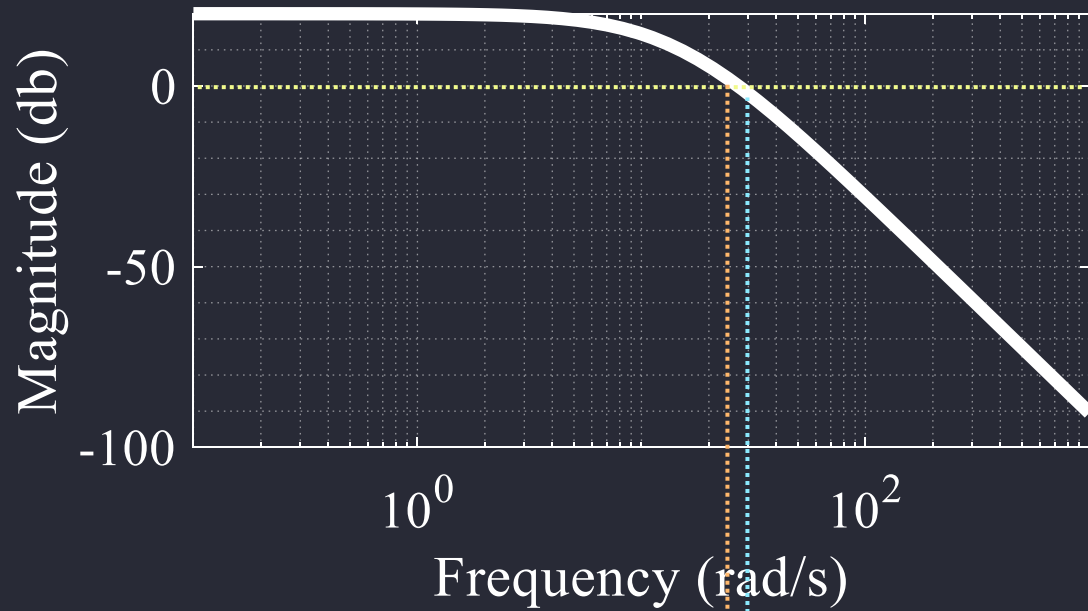
$$pm = 0^\circ$$

Marginalmente Estable

$$\text{Aumento Ganancia: } k_{gm} = 4.375(1) = 4.375$$



$$0 < k < 4.375$$

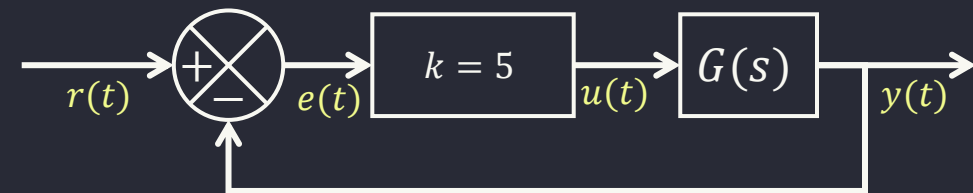


$$gm = 0.875 \text{ (Abs)}$$

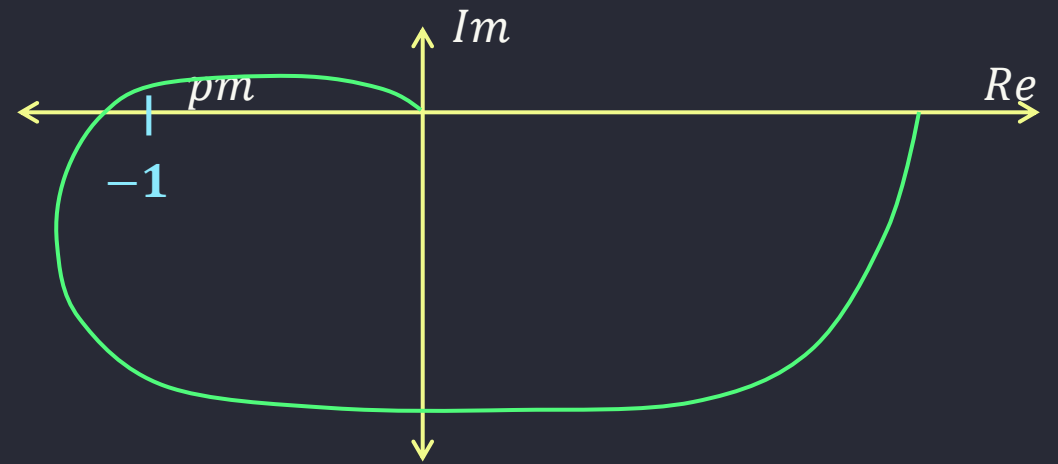
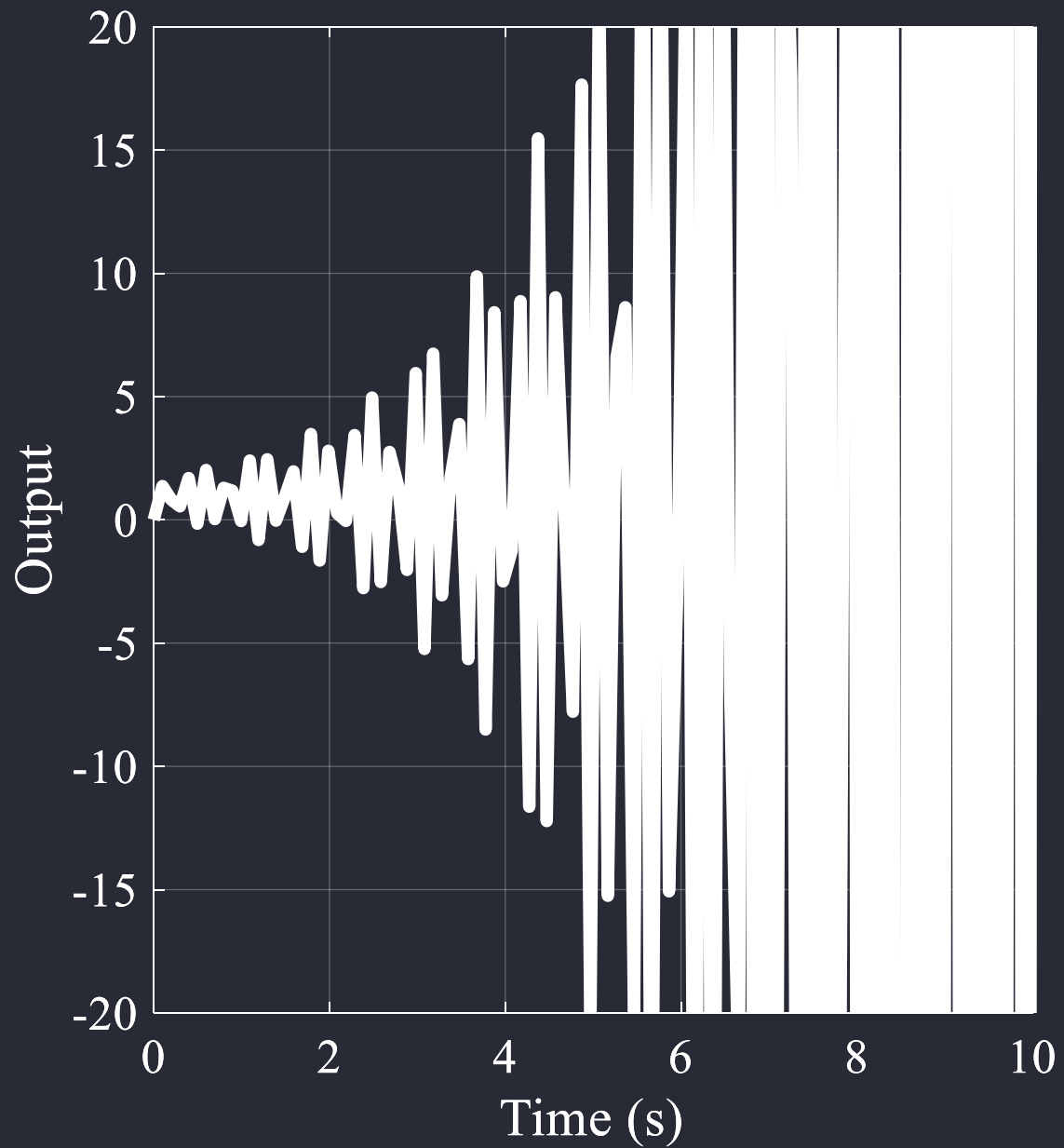
$$gm = 20 \log(0.875) = -1.16 \text{ dB}$$

$$pm = -4.22^\circ$$

Inestable



$$0 < k < 4.375$$

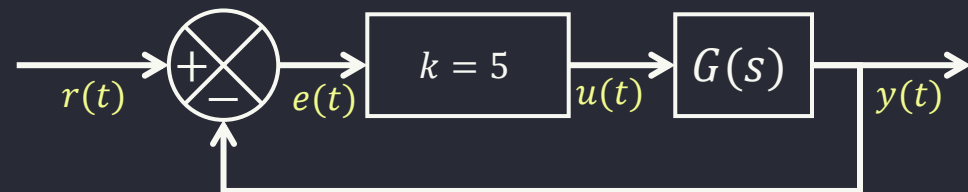


$$gm = 0.875 \text{ (Abs)}$$

$$gm = 20\text{Log}(0.875) = -1.16\text{dB}$$

$$pm = -4.22^\circ$$

Inestable



$$0 < k < 4.375$$